The Fable Computer, Part III: A Ternary Language Model on the Regenerative Graphene-Plasmon Fabric

Embedding Ternary Bonsai 2 27B as a streaming matrix-multiply coprocessor — a digital lane on the Part-I logic fabric, bit-exact as a netlist, a cold analog counter tree on the Part-II interference fabric, and the one measurement that decides both

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Authorship and Status Disclaimer. This manuscript extends the author's design studies [1, 2] ("Part I", "Part II" throughout). Generation of content involved the use of an AI, Anthropic's Claude, at the author's direction: the two design studies folded into Sections 3–7 were executed in a Claude Code session (model Claude Opus 5.5) and each was re-checked by an independent adversarial verification agent before its second revision; the manuscript itself was drafted on Claude Opus 5.5 under orchestration by Claude Fable 5.1 and revised after an AI proofreading pass. Its account of the community record promoted through 17 October 2026, and a re-check of its external facts against primary sources on 24–25 October 2026, were prepared in a maintainer-operated Claude Code session in which Claude Fable 5.1 orchestrated Claude Opus 5.5 agents for mapping, two-lens adversarial verification, drafting and checking. The community-record notes it relies on were themselves produced and assessed by the project's disclosed agent pipeline, each note's header giving its promotion route and, where the note records them, its executing models (some were promoted by maintainer direction after the pipeline's four-round cap). The author declares a lack of formal expertise in machine-learning accelerator architecture and in graphene-plasmonic device physics, and the document has not received independent expert review. All physical claims are reduced-order model results in the sense of Parts I and II ("in-model": established within the released chains and the named extensions, under stated assumptions, not measured) and must be treated as a starting point for discussion, not as peer-reviewed engineering guidance.

Abstract

A ternary-weight language model is a natural workload for the regenerative graphene-plasmon fabric of Parts I and II: its inner loop is "add, subtract or skip an 8-bit number", and the fabric's primitive, in Part I's design, is a threshold cell that adds and restores (in-model; the released cell compresses rather than thresholds and re-clamps a loaded cascade near a 2 % drive, not at the coded rail; notes 2026-10-08, 2026-10-16; the dated notes cited throughout are files in the repository's notes/ directory, https://github.com/ryoji-info/FableComputer/tree/main/notes, listed with their file names in reference [12]). Ternary Bonsai 2 27B [3–5] has 27 B weights in {−1, 0, +1}; one decoded token takes 25.6 G ternary multiply-accumulates (MACs) and streams 5.67 GB of weights (external). The embedding is a streaming coprocessor: a conventional (CMOS) host keeps every stateful and floating-point task, and fabric lanes — fixed datapaths of cells — each compute one 32-wide block sum of 8-bit activations and ternary weights per 14.5-ps slot, assuming the llama.cpp family's Q8_0 activation blocks, which PrismML's own runtimes use on only some of their paths (external). The digital lane is bit-exact (demonstrated (runnable) on 1.70 M Bonsai-shaped block sums; a netlist check, not physics) and, timed for latch-free operation, takes 8,169–8,455 cells, 0.22–0.23 mm^2 and 64–86 mW, or 29–39 fJ per MAC, at 300 K at the hydrodynamic tier (in-model; no kinetic correction is applied at 300 K, and every correction the record evaluates would raise the dissipation; notes 2026-07-13, 2026-10-15). Two, three and seven lanes keep pace with 0.5, 1 and 3 TB/s (88, 176 and 529 tok/s at batch 1) in at most about 1.7 mm^2 and 0.64 W (in-model); the system stays memory- and I/O-bound. Against the error that Q8_0 quantization adds to the activations, a lane may err at most 6.3×10^{-9} per cell decision (in-model; an upper bound). Only the released booking of that error passes warm, and only on the coded rail swing of 3,833 quanta,

which no run of the record delivers to a deciding input; at the swings the record's own loaded cascade sustains (99–277 quanta at fan-out 2 on the absolute and ratio bias legs; no multi-cell run) no booking passes warm, and at its central fixed point none passes at any temperature (in-model; note 2026-10-08). On the rail swing, the split and junction losses of Part I's fan-out budget (Part I §6.3) lower the pass temperature to 144 K for its −3 dB split alone and to 114 K through a −1 dB junction (colder at −3 dB), and the operand launch must be 28–33 dB quiet against the record's coded 20 dB (in-model). On the Part-II interference fabric, a three-input interference decoded by three comparators (QFA-3) can replace 184, 154, 65 and 15 of the Wallace tree's 236 full adders at 4, 20, 77 and 300 K while the dot product keeps 41 dB, 1.5 dB below the digital lane's 42.5-dB pass line (in-model, with inputs relaunched from the coded rail level, which no run delivers; five rework items open, one of which estimates these Student-t SNRs 1.7 dB optimistic), saving at most about a third of the cells ($\leq$ 32 %) even if every counter were analog; a 77 K lane wins under booking A0 only with a cooler at about 40 % of Carnot or better (39.6 % for the hybrid, 41.4 % all-digital), and 20 K and 4 K lanes lose even at Carnot (in-model; cold-cell dissipation uncertified). Both routes end at one open measurement, the loaded decision error of a cell versus temperature, reported with the swing its deciding input receives (bench gate DG1); the digital lane is recommended for Bonsai, and QFA-3 is kept as a test tile. Of the levers (in-model, on synthetic activations), 4-bit activations would raise the split-and-junction (A2–A3) pass temperatures on the rail swing from 114–144 K to 168–213 K, while whether a model keeps its quality at 4-bit activations is external (near parity has been reported with quantization-aware training, but not for post-training quantization to ternary weights; Section 8.2), and a threshold full adder would halve lane logic on a cell that thresholds a three-way sum and takes a depleting input; in the amplitude encoding the released cell separates two in-phase inputs from one by only 2.6–3.3 dB and has no depleting input for a carrier-frequency logic signal (in-model; a three-way sum is untested; note 2026-10-16). Nothing has been built; the gain cell is experimentally unproven, and whether graphene's own electron fluid has the drift threshold the cell is biased against at 353 K is open (note 2026-10-15).

1. Introduction: a model that only adds, and a fabric that only adds

A large language model generates text one token at a time. Each decode step pushes one activation vector through every layer, and almost all of its arithmetic is matrix–vector products: every weight is read once, multiplied by one activation and added into an output. When one user is served at a time (batch 1), each weight fetched from memory is used once, so the processor waits on memory, and the decode rate is the memory bandwidth divided by the bytes of weights read per token — the memory-bound corner of the roofline model [6, 7] (external).

Ternary models change the arithmetic, not that bound. The BitNet b1.58 line of work [8], which grew out of the binary-weight BitNet [9], trains networks whose weights are −1, 0 or +1, with one floating-point scale per tensor (per group of 128 in Bonsai), while activations stay at 8 or 16 bits: a multiply becomes "add", "subtract" or "skip", and a weight costs about 1.7 bits instead of 16. Ternary Bonsai 2 27B [3–5], derived from Qwen3.8-27B [10], is an openly released instance at a scale where memory bandwidth dominates; it keeps 98.2 % of its parent's benchmark average (external; Section 3).

The Fable Computer is a machine design on a graphene-plasmon fabric (Section 2). In Part I [1] every gate is a graphene cavity about half a micrometre long, biased by a DC current just below self-oscillation, so that it amplifies a terahertz plasmon pulse and, in Part I's design, clamps it against a saturating level, the rail; logic is a threshold on the sum of input fields. In-model the released cell departs from that design (Section 2.1): it has no output plateau, its soft compression re-clamping a loaded cascade near a 2 % drive instead (note 2026-10-08); it has no threshold AND, two in-phase inputs exceeding one by only 2.6–3.3 dB; and it has no density-route NOT, inverting in the amplitude encoding only by interference or behind an external electrical gain stage (note 2026-10-16). Under phase encoding a NOT

is a half-period delay, with restoration from a pump at twice the carrier frequency admissible in-model only in a narrow window (note 2026-10-17). The fabric runs wave-pipelined — operations follow each other through the cells one clock slot apart, with no latches — and stores nothing. Part II [2] showed that below the rail the same fabric is a quantum-limited analog computer whose interference is decoded to one or two bits by threshold cells of the same design. The gain cell itself has never been demonstrated.

The workload and the fabric fit. At batch 1 the weights stream from DRAM on every token anyway, so the fabric's lack of storage forfeits no weight reuse that a batch-1 GPU would have had; what it does cost is that every activation bit must be relaunched in the slot that uses it, which makes the operand lines the hardest engineering in the design (Section 4.6). The fabric cannot decode faster than the memory allows, so the questions are what maps onto it exactly, whether that piece belongs on Part I's restoring logic or on Part II's interference, and which unmade measurement decides whether it works.

In brief: a sum of 32 products of 8-bit activations and ternary weights per slot — the Q8_0 activation block of the llama.cpp family [11], adopted here as an assumption — fits on a Part-I lane, a fixed datapath of cells, that is bit-exact as a netlist and needs about 8,200–8,500 cells; two to seven lanes keep pace with 0.5–3 TB/s (Section 4). Whether the lane is also correct depends on how often a cell decision flips: the record's pieces admit five bookings, spanning seven orders of magnitude in lane error power at room temperature (lane signal-to-noise ratio, SNR, from 64.9 to −4.0 dB; Table T7), and only the released one passes, on the coded rail swing (Section 5). Part II's interference counters help only when cold, saving at most about a third of the cells, while the cooler costs more unless it is a good one (Section 6). Both routes end at bench gate DG1, a pre-registered pass/fail measurement of a loaded cell's decision error versus temperature, reported with the swing at the deciding cell's input (Section 7).

Section 2 summarizes Parts I and II (Tables T1, T2); Sections 3–8 develop the design; Section 9 states what is demonstrated, in-model, external and open and lists the limitations; Section 10 concludes; the appendices document the listings and the symbols. The labels of Parts I and II are kept: **demonstrated** — executed against the released, unedited chains (for a netlist identity, "demonstrated (runnable); a netlist check, not physics"); **in-model** — established within the record's models plus this paper's named extensions, under stated assumptions, not measured; **open**; and **external**, for facts about Bonsai, GPUs and memory. "The record" is the two released model chains plus the community notes promoted into the project repository [12], cited by date; each note is the file in the repository's notes/ directory whose name begins with that date (reference [12] gives the file names; notes/INDEX.md lists what each note settles and what later notes did to it, and each note's header gives its promotion route). Open design questions and measurement requests filed as repository issues [25] are cited as "issue #139" and the like, as open items, never as results.

2. The fabric in brief (what a reader needs from Parts I and II)

This section is for readers new to Parts I and II. Every fabric term is defined at first use, and the principal ones are collected in Table T1; Section 2.5 states what Part III keeps, changes and considered instead.

2.1 One cell, every gate (Part I)

The cell is a Dyakonov–Shur (DS) cell: a 576-nm cavity of graphene encapsulated in hexagonal boron nitride, at a carrier density $n = 10^{12}$ cm^{-2} under a metal gate, resonant at 1 THz for the gate-screened (acoustic) plasmon. A DC drift current flows through it; Dyakonov and Shur showed that such a channel, with electrically different ends, self-oscillates above a threshold drift. Part I biases the cell at 0.7 of that threshold (a tracked ratio, recalibrated in situ), where it amplifies instead: an injected pulse recirculates and leaves stronger. This regenerative gain is +9.7 dB for a continuous wave (CW; the analytic enhancement ratio) and about +8 dB per pulse in a settled 0.25-THz train (the released solver's per-slot

peak in its shipped cavity; a design-length rerun measures +9.2 dB). At that 4-ps slot Part I §7.3 splits the +8 dB into ≈+3.0 dB (band 2.9–4.0) of data-independent gain and a ≈+4.8 dB predecessor pedestal, the same accumulation it books as inter-symbol interference, so the two cannot both be banked; at this Part's gated slot a duty-0.8 burst measures a worst-case-'1' gain of 7.7–7.9 dB (on a 720-point solver grid at 67–71 GHz, the cell at 0.7 × the analytic threshold; note 2026-07-31; Section 2.2). These figures are in-model at the hydrodynamic tier (the record's fluid model of the electrons), with a signed −0.95 dB per cell from the kinetic closure at 353 K still to be certified, plus a separate, still-open ±1 dB allowance for the clamped contacts that could be larger for a diffuse contact, one that randomizes the momentum of the electrons it reflects, where a specular one reflects them mirror-like (Part I §7.3, §7.5; note 2026-07-13).

The kinetic figure presupposes a threshold to track, and whether graphene's own electron fluid has one at 353 K is open. Its drift law alone keeps one, at ×1.7 the record's threshold drift, for a cold fluid under the released friction law with the dissipated energy removed; but the single-rate finite-temperature kinetic operator that note 2026-07-13 §6 specifies finds none at 353 K below the Landau ceiling (the drift at which the cavity's slow, backward-running leg falls to the Fermi velocity) at the central and lower electron–electron rates of that note's band when the electrons shed their drift heat. What decides is the relaxation of the current not locked to momentum — its rate, which the record does not pin, and its finite-drift structure — and three admissible two-rate continuations, identical at zero drift, give three answers (in-model; note 2026-10-15).

In Part I's design the pulse saturates against the rail, so every cell restores logic levels. The released (unpumped) model has no output plateau below near-depletion — demonstrated through a 70 % drive on the absolute and ratio bias legs (the solver's cell held at 0.7 × the coarsest grid's threshold, or at 0.7 × each grid's own), though the model leaves its physical domain above a few-percent drive, where its fluid outruns graphene's Fermi velocity — and its soft compression re-clamps the '1' amplitude gate after gate at a loaded fixed point near a 2 % drive (99–277 quanta at fan-out 2), not at the rail (in-model, under note 2026-08-01's cascade assumptions, with no multi-cell run; note 2026-10-08).

Logic uses the same transfer. In Part I's design model (its Eq. (5)) a cell fires when the summed input fields on its gate cross its threshold — a high threshold makes AND, a low one OR, and a depleting coupling (pushing the cell below threshold when the input is high) makes NOT. On the released cell neither the threshold AND nor the depleting NOT exists (in-model; note 2026-10-16): its only density coupling, rectification of a carrier-frequency input, accumulates — a delivered '1' raises the drain density by +0.86–1.06 of its swing on the retuned absolute bias leg — and two in-phase '1's exceed one by only 2.6–3.3 dB, so the cell compresses rather than thresholds (an OR still fires on either input, with no margin). In the amplitude encoding the only in-model inverters are destructive interference of the input with a coherent clock pulse, ≥ 10 dB only within 10.1–12.3° of phase error (by grid, tightening under refinement), and a detector segment behind an external electrical gain stage; under phase encoding NOT is a half-period delay, with restoration from a pump at twice the 1-THz carrier frequency f_0 (a $2f_0$ pump) admissible in-model only in a narrow window (note 2026-10-17; Section 8.1(g)). The steepness is the sharpness $k = \beta \cdot A_{swing}/\Delta n$ (coupling times logic swing over threshold width); the design rule $k \geq 8$, with a 10-dB on/off extinction, gives a static noise margin — the input disturbance a restoring cell rejects — of 0.256 of the swing (the released code's closed-form value, which never calls the cell's transfer; notes 2026-07-21, 2026-08-02; Part I §5 quotes ≈0.2). It is a design rule, not a property of the released cell: on Eq. (5)'s tanh scale the cell's density coupling gives $k = 0.18–0.26$ at the delivered swing and its interferometric inverter ≈ 2.9, so no released route supplies that margin (in-model; note 2026-10-16). There is no multiplier and no native plasmonic storage: "compute in plasmons, store as charge" (Part I §6.4). Persistent bits live as gate charge in conventional electronics. On the fabric Part I allows only small fast registers of pumped bistable cells or recirculating loops (Part I §6.4), and

among its optional room-temperature terahertz electronics it names resonant-tunnelling-diode (RTD) latches as fast electronic registers (external devices, Part I §9.8); the released chains model none of these as a register — note 2026-10-17 finds, in-model, that a $2f_0$-pumped released cell latches a bit under a CW pump but, gated in the slot, re-levels no flip (Section 8.1(d)) — and this Part uses none, so its lane has no latch.

Part I's half adder is an OR, a carry AND, a NOT, a delay buffer (DLY) and an output AND — five active cells of depth 3 — plus two optional carry-alignment buffers: seven biased cells (Part I §6.5). The DLY cell only equalizes path lengths so that both inputs of the output AND arrive in the same slot, a discipline Section 4.3 generalizes. Part I composes a full adder from "two half adders and an OR": 2 × 7 + 1 = 15 biased cells of depth 7, this Part's reference.

2.2 Timing, fan-out, stations

Part I's clock is optical: a microcomb-photomixing chain is to launch phase-stable pulse trains at the fabric edge (Part I §3, §6.2). The demonstrated chain emits a continuous wave, so the pulsed clock is a proposed extension of demonstrated hardware (Part I §3.2), and a primary-source check found no chip-scale source of the gated burst this Part launches (external; Section 8.1(e)). A launch is the injection of a clock or data pulse; a slot is one clock period. Operation is wave-pipelined [13], so timing rests on balanced path lengths, and re-sync stations — DS cells used as clocked ANDs, fed by optical clock taps — periodically re-time pulses and flush noise. Part I wrote "every 10–24 gates"; the released spacing law gives 13.3 cells at the cell's noise-figure floor of 2.77 dB (NF, the factor by which a cell degrades the signal-to-noise ratio) and 10.0 cells at Part I's NF ≤ 4 dB requirement, and no documented margin convention reaches 24: at the law's 6-dB margin 24 cells would need NF = 0.198 dB, below the 2.77-dB floor (note 2026-08-25 §5 item 5; digital_lane_bonsai.py §0). NF = 4 dB is a requirement, not a value of the released cell law: with its noise factor $2 - 1/G$ a bare cell's NF stays below 3.01 dB at any gain $G \geq 1$ (in-model), so the 10-cell end is reached only through what precedes the gain — a blind flush of ≥ 1.23 dB ahead of it takes the NF past 4 dB, an in-model upper bound — or through an amplifier idler hotter than the released booking, whose added-noise charge is itself undetermined in sign and magnitude (note 2026-08-05). This Part uses 10–13 cells, from Part I's requirement (10.0) to the released floor (13.3).

Which compact source should supply the module's clock — the photomixing chain, a resonant-tunnelling diode, a silicon harmonic source or a self-oscillating cell — and how it is distributed (power per launch point, number of launch points, master-to-station loss, a burst modulator with ≈1-ps edges) is an open design question (issue #139; Section 8.1(e)). What the launch must meet the record does fix: the slot-confined, data-gated duty-0.8 burst described below (note 2026-07-31; in-model), a launch SNR of 28–33 dB (Section 5.4; in-model) and, at an interference inverter, a clock pulse matched to its input within the phase window of Section 2.1 and within ±21 % (coarsest grid) to ±17.6 % (finest grid) in amplitude (note 2026-10-16; demonstrated on the released solver). The comb fixes the carrier phase at the launch; each cell then adds its own transfer phase, 12–21° per 1 % of drive detuning and 4–6° per bias step of $0.05 \cdot M_{th}$ (in-model; note 2026-10-13), against Part I's ±15° clock budget, and how much of that budget the lane can spend turns on two open forks: whether its cells run in the clocked regime of Part I's Eq. (5), each amplifying the pump, or as a station-free cascade, and whether a two-input gate sums fields or couples envelopes (Section 9, limitation 31).

Fan-out F costs 3 dB per branch of a Y-split plus junction loss; a junction is a gate-defined step between cells, budgeted at −1 to −3 dB (Part I §6.3, §9.4). Only F = 2 closes: note 2026-07-21 gives an F = 2 reserve of +2.01 dB (band 1.46–2.66, near-specular CW) at the 0-dB junction class, computed without junction attenuation (Part I §6.3), an effective sharpness $k_{eff} = 8 \cdot F^{(-½)} = 5.66$ (static noise margin 0.221 at the same 10-dB extinction) and a disorder ceiling F ≤ 2.56, the largest fan-out at which k_{eff}

stays above the regeneration floor k = 5 (in-model). The reserve is a CW small-signal figure. At the gated slot described below the existence headroom of a duty-0.8 launch — the small-signal gated gain less the loaded-cascade F = 2 requirement of 6.515 dB (0.95 dB of kinetic correction, the 3-dB split and the per-gate loss, with no compression term) — is +1.21 dB at 67 GHz and +0.98 dB at 71 GHz on a 480-point solver grid, with the cell at 0.7 × the analytic threshold, near-specular contact and the kinetic bulk term at its central value, and it turns negative at 71 and 77 GHz at the −1.00 dB contact corner (note 2026-08-01, gated requirement round trip; demonstrated rows, in-model composition); the two drive classes are not commensurable. The $k_0 = 8$ behind k_eff, its margin and the ceiling is the released code's design constant (cell.py), not a property of the released cell (Section 2.1), so the disorder ceiling applies to no released cell; that F = 2 closes rests on the gain reserve, which this qualification leaves untouched (in-model; note 2026-10-16), and, like every figure at the tracked bias, presupposes the threshold of Section 2.1 (note 2026-10-15). Wider fan-out uses trees of F = 2 buffers.

The slot rate has moved since Part I's 4-ps (0.25 THz) baseline. Because a regenerative cavity rings for a few picoseconds after a pulse, each slot must be flushed, and F = 2 logic then closes only near 0.1 THz (note 2026-07-23); with a physical launch confined to its slot and gated by the data — a duty-0.8 burst of the 1-THz carrier — the F = 2 frontier is 69 GHz (central value; band 63–74 GHz; 67–71 GHz at the demonstrated rungs, the 71-GHz edge only under the round floor specification below), adjudicated at 353 K with the cell biased at 0.7 × the analytic threshold (note 2026-07-31; in-model composition), not at 0.7 × the solver's own grid threshold, the ratio in which Part I quotes solver results. Three notes of 2026-08-01 qualify it. The gain-requirement check measures the gated drive's compression knee in the requirement ledger's own convention and, with the ledger's one imported operating amplitude still in it, puts the F = 2 requirement at that amplitude, the compression term included, at 7.0 dB on the released grid and ≈7.3 dB extrapolated (demonstrated at each grid; the extrapolant in-model); it confirms the 67–71 GHz window without extending it, but its requirement leg sat at the ratio bias and its gain rows at the analytic one, a fork worth +1.20 dB of gain and 3.5 dB of floor on one 77-GHz row (gated requirement round trip). With both legs at the ratio bias the bench will use (0.7 × the measured threshold; matching it in the solver with 0.7 × the solver's grid threshold is the reissued G1's proposal), the gain clears that requirement by +1.1 to +1.3 dB at 67–71 GHz (demonstrated as arithmetic on two measured legs, in-model as a margin; ratio-bias gain table), and note 2026-08-02 adds the 77-GHz knee, cleared by +1.04 dB on the coarsest grid. The reissued bench gate G1, a proposal, keeps its CW existence test (G1a) and adds a gated-drive knee report (G1b).

At the ratio bias the '0'-slot floor binds. In the released solver this floor is the largest intracavity velocity perturbation in a '0' slot, referred to the mean '1' — a per-slot peak that, in a diagnostic at 67 GHz on the coarsest grid, sits near the cell's source end, not at its drain (note 2026-08-02; demonstrated). Under the strict specification (−11.8 dB, $20 \cdot \log_{10}$ of the released noise-margin formula's 0.256, whose input is the round 10-dB extinction, so the two are one chained assumption) no gated slot rate passes robustly; under the round ≥ 10 dB specification 67 GHz is robust and 71 GHz marginal (note 2026-08-01, ratio-bias gain table). At 67 GHz on the coarsest grid the strict margin, +0.007 dB in the released velocity reading, reads −0.039 dB at a second carrier phase and −0.048 dB in density, and on a 480-point grid density moves it the other way, so the strict test is decided at a precision the observable does not define (note 2026-08-02; demonstrated). The same note, revised and re-submitted, finds that the released code's compact logic map (cell.transfer, which is not Part I's Eq. (5) and which nothing in the released pipeline calls) cannot fix the floor specification — composed as an inverting pair it supplies attracting extinctions from −0.04 to at least −11.69 dB, set by constants the chain never fixes — so the specification must be imported, from the bench or from a downstream decode demand that is itself set by conventions the released code leaves open (comparator re-trim, phase projection, k_dec and c; in-model; notes 2026-08-02, 2026-08-12). At its fitted convergence order note 2026-08-02 resolves 71 GHz to a

0.31-dB fail of the round specification while 67 GHz clears it at −10.88 dB (in-model and ill-conditioned: the sign of the correction is robust, its size is not, and no fourth grid rung of the floor has been run). At 67 GHz the slot-mode amplitude a decision consumes sits 3.83 dB below that per-slot peak (3.89 dB in the drain-plane density), and only the '0' slot right after a '1' carries a binding residue, −15.9 to −15.7 dB below the mean '1' and 82–86° off its phase on the released grid and time step (−15.0 dB at 65–69° on a 720-point grid; later '0' slots sit ≥ 21 dB lower; note 2026-08-12; demonstrated at the model's ratio bias). That phase moves with the grid and no continuum value is registered (note 2026-08-13), and the residue moves 10.3 dB as the solver's time-step factor runs 0.2–0.8; the specification names neither denomination (per-slot peak or slot mode), so none of this moves a floor verdict. The frontier, 69 GHz (band 63–74 GHz), is neither confirmed nor moved (open). This paper uses 69 GHz (a 14.5-ps slot); every lane count, tok/s per lane and line rate below scales with the slot rate.

The released chain stores a cell dissipation of 12.67 μW at the 353 K (80 °C) cap, with an upper bound of 16.64 μW (1.267 and 1.664 kW/cm^2 over 10^{-8} cm^2, hydrodynamic tier). Scaled to 300 K as τ^{-3} at the tracked bias (τ the plasmon lifetime; extension [D3] — bracketed labels [D…], [E…] and [V…] name this paper's in-model extensions, defined in Table TA3) it is 7.78–10.22 μW; kinetic-corrected at 353 K it is 15.9–16.7 μW (digital_lane_bonsai.py §0). A cell is charged about 11 sites of Part I's gate lattice (27.4 μm^2, the half-adder block's 12 × 16 μm over its 7 biased cells; Part I Table 5), of which its active area is 1 μm^2, so cells fill 3.6 % of a lane. The chip may rise 55 K above ambient; with the kinetic correction Part I's margin at a one-third fill (the area fraction under biased cells) is about 4 % (Part I §8.2). The kinetic-corrected figures are the closure's (Part I §7.5). In graphene's own electron fluid, wherever a threshold exists, tracking 0.7 of it costs ×1.2–2.1 (hydrodynamic rows, 300 and 353 K) or ×3.2 (the one kinetic row with a threshold, at 300 K) the stored 353 K cell dissipation in the released Joule form, and ×1.25–1.35 more in graphene's own; under graphene's form the rise at Part I's one-third fill is 61–176 K against the 55-K budget; under the released form one row passes, at 49 K (300 K); and at 353 K the single-rate kinetic operator of note 2026-07-13 §6 has no tracked bias at the central and lower electron–electron rates of that note's band when the electrons shed their drift heat, and at the band's top none only under the note's projected treatment of the DC drive (dropping that term gives a crossing; in-model; note 2026-10-15). Every number in this paragraph, and every gain above, rests on Part I's lifetime row (τ = 1.0 ps at 300 K, 0.85 ps at 353 K), which Part I states as a best-case assumption in premium-material territory, against a measured room-temperature baseline for propagating acoustic terahertz plasmons of $\tau \approx 0.4$–0.6 ps ($Q \approx 2.5$–3.8; Part I §4); whether a scalable graphene/hBN stack reaches the row is an open design question (issue #140).

2.3 The quantum view (Part II)

Part II counts pulses in quanta. One plasmon of the 1-THz cell mode is a 0.16 % density swing; the knee (the 1-dB compression point, where the linear window ends) is ≈38 quanta at the launch plane (band 22–51 on the released solver grid, lower on finer grids, by an amount that depends on the bias leg; a reference plane is the point along a pulse's path at which an amplitude is quoted), and the rail ≈3,833 quanta, anchored at no plane (Part II §3.1; notes 2026-07-17, 2026-10-08) — classical solver amplitudes converted to quanta. At the drive plane a 10 % swing is, in the released solver, a 9–14 dB gain-compression point with no output plateau, depending on bias leg and observable, and neither the drive plane nor the intracavity plane holds a clamp (demonstrated for the released model, whose fluid outruns graphene's Fermi velocity above a ≈0.03 drive; note 2026-10-08). With $T_Q = \hbar\omega_0/k_B \approx 48$ K, the mode carries $\bar{n} = 5.8$ thermal quanta at 300 K and about one at 77 K. Part II excludes quantum logic on this cell and builds a quantum-limited analog computer under two rules: gain only after a decision, and a quiet launch.

Its unit QMAC-1 interferes two pulses of N_op quanta in one junction and decodes the sum with two comparator cells — Part I's half adder re-programmed, except that the comparators need $k \geq 16$, a redesign (Part II §5). Its 2-bit decode errs 0.14 per shot at 300 K, its 1-bit decode 3.9×10^{-3}, and a rail-restored digital decision 3.6×10^{-11} (Part II Table Q3). This paper calls that last figure booking A0, a booking being a stated convention for turning a decision's swing and noise into an error rate (Section 5.3); A0 holds at the coded rail swing of 3,833 quanta, which no deciding input in the record carries (in-model; note 2026-10-08). Part II's Table Q3 values assume an amplifier idler at the lattice temperature; for an idler no colder than the lattice they are optimistic-or-exact on that axis, with a lever of opposite sign on the gain booking unsettled (note 2026-08-05).

Three Part-II quantities are open. The launch: Part I's re-sync spacing law consumes a 20-dB launch SNR against the thermal floor (Part I §6.2, §7.4), and Part II codes it as the launcher's own 10 % amplitude jitter — two different quantities, both present in the released artifacts; which one Part I intends, at which plane, is open (note 2026-08-25). In Part II's coding, at N_op = 400, the 2-bit decode cannot be cooled below 3.4×10^{-3} (Part II §4.2), and a 10^{-3} class exists only for a jitter of at most 8.16 % (a launch SNR of at least 21.76 dB; in-model; note 2026-08-25). N_op: the default 400 quanta per input is called "launch-budget-capped", a cap the record never derives: no released key or manuscript sentence defines what it denominates, and 400 quanta is 18.41 dB over the thermal occupation at 300 K counted in quanta but 21.06 dB in quadrature, so whether it sits below or above a 20-dB budget depends on the convention (note 2026-08-25). And c, the comparator's rms trim offset as a fraction of its transition width, is coded at $1/\sqrt{12} = 0.289$ and derived nowhere (note 2026-07-15).

2.4 What is demonstrated

Nothing physical. No DS cell has been built and biased as a regenerative amplifier; Part I's bench gates G1–G5 (G1: single-cell net gain at ≤ 80 °C) and Part II's QG1–QG5 are unrun, and the one current-driven graphene terahertz plasmonic amplification reported as of June 2026 (Part I §2) is attributed by its authors to another mechanism. The record is the two released chains (fable-model-chain/, fable-model-quantum/), executed unedited, and the promoted notes [12]; every physical number below inherits the cell's in-model status.

Part I also places a data gate on this Part: a near-model sub-threshold cavity linewidth on a specular-contact cell, measured with a contact-preparation contrast, a temperature sweep and a drift sweep, "gates an architecture-led Part III on data" (Part I §10), because the open ±1 dB contact allowance dominates the per-cell gain budget behind the F = 2 reserve and the bookings of Section 5 (Part I §7.5; notes 2026-07-18). That measurement is unmade too. Part I gives it no numeric pass threshold (note 2026-07-18 pre-registers its bands); its protocol is posted as a measurement request (issue #88), the reissued G1, a proposal, places it as the rung before the gain test (note 2026-08-01), and the staged bench plan, an open call for an experimental partner (issue #97), runs it first because it needs no gain. Once the contact channel is separated, its temperature sweep and its joint reading of the line's width and shift would also bear on the open electron–electron rate behind the cell's 353 K threshold (discriminants, not falsifiers; in-model; note 2026-10-15). It stands in Table T10 as the prerequisite L0, beside G1.

Term	Meaning	Value at the operating point	Where
DS cell	576-nm gated-graphene cavity with DC drift, biased below self-oscillation; every gate is one	$n = 10^{12}$ cm^{-2}, 1 THz, bias M/M_th = 0.7 (tracked)	Part I §5

Term	Meaning	Value at the operating point	Where
Regenerative gain	multi-pass amplification of an injected pulse in a sub-threshold cell	+9.7 dB CW (analytic); ≈+8 dB per pulse streaming (shipped cavity, per-slot peak; +9.2 dB at design length; ≈+3.0 dB of it data-independent at 4 ps); gated worst-case '1' 7.7–7.9 dB (720-point grid; 0.7 × the analytic threshold); kinetic −0.95 dB/cell at 353 K (within the closure; threshold existence open)	Part I §7.3, §7.5; notes 07-31, 10-15
Rail	Part I's saturation level that re-clamps a '1' at every cell, coded as the '1' swing (EPS_RAIL) that every booking (A0–A3, B), [D8], [E7] and the QFA-3 caps assume; the released (unpumped) cell has none below near-depletion — at the drive plane the rail is a 9–14 dB compression point, with no output plateau through a 70 % drive (demonstrated in the released model, which leaves its physical domain above a few-percent drive) — and in-model the loaded cascade re-clamps near a 2 % drive instead	≈10 % density swing ≈ 3,833 quanta, anchored at no plane; delivered to no deciding input by any run	Part I §5; Part II §3.1; notes 07-17, 10-08
Knee	1-dB gain-compression point; end of the linear window	≈1 % swing ≈ 38 quanta at the launch plane (22–51 on the released grid; lower on finer grids)	Part II §3.1; notes 07-17, 10-08
Threshold logic	Part I's design: AND/OR = threshold on summed input fields; NOT = depleting coupling	high threshold AND, low threshold OR; released cell (in-model): no threshold AND (2.6–3.3 dB), no density-route NOT; NOT by interference or behind an external gain stage in the amplitude encoding, by a half-period delay under phase encoding	Part I §5; notes 10-16, 10-17
Sharpness k	threshold steepness $\beta \cdot A_{swing}/\Delta n$; sets the static noise margin (NM)	$k \geq 8$ (10-dB extinction): NM 0.256; $k_{eff} = 5.66$ at F = 2: NM 0.221 (design rule); released cell 0.18–0.26 by density coupling, ≈ 2.9 by interference (tanh scale)	Part I §5; notes 07-21, 08-02, 10-16

Term	Meaning	Value at the operating point	Where
Fan-out F	cells driven by one output through Y-splits (−3 dB per branch)	F = 2 (CW reserve +2.01 dB; gated headroom +0.98–1.21 dB at 0.7 × the analytic threshold; ceiling 2.56 through the design constant $k_0 = 8$)	Part I §6.3; notes 07-21, 08-01, 10-16
Junction	gate-defined step between adjacent cells	−1 to −3 dB	Part I §9.4
Slot	one clock period; one operation per pipeline per slot	14.5 ps at 69 GHz (band 63–74 GHz; 0.7 × the analytic threshold); open in the bench convention, where the floor specification must be imported	notes 07-23, 07-31, 08-01, 08-02
Launch	injection of a clock or data pulse at the fabric edge	Part I: a 20-dB launch SNR against the thermal floor; Part II: coded as 10 % amplitude jitter; two different quantities	Part I §6.2, §7.4; Part II §4.2; note 08-25
Wave pipelining	operations one slot apart with no latches; timing by path balance [13]	—	Part I §6.5
Re-sync station	clocked AND fed by an optical clock tap; re-times and flushes noise	every 10–13 cells (NF 4 dB, Part I's requirement, to the 2.77-dB floor)	Part I §6.2, §9.6; notes 08-05, 08-25
Quantum, N	one plasmon of the cell mode: a 0.16 % density swing	$\hbar\omega_0 = 4.14$ meV; T_Q = 48 K; $\bar{n} = 5.8$ (300 K)	Part II §3
N_op	quanta per input pulse at logic 1 in an analog tile	400 registered (a cap never derived, note 08-25); 2,600 in this paper's REF (relaunched from a rail-level '1', §6.3)	Part II §5; §6.3
QMAC-1	two-input interference decoded to 2 bits by $k \geq 16$ comparators	2-bit error 0.14 (300 K), 3.1×10^{-3} (77 K); quiet launch, idler at the lattice	Part II §5–6
c	comparator rms trim offset, fraction of its transition width	coded $1/\sqrt{12} = 0.289$; open	Part II §5; note 07-15
P1, P2	rail reference planes: the pre-loss launch sum; the combiner output	both carried; the record anchors the rail at neither	§6.2–6.3
REF, CONS, PERM	the analog tile's three conventions: reference (rail at P2, relaunched inputs, $c = 1/\sqrt{12}$); conservative (rail at P1); permissive (quiet launch, c = 0.02)	N_op 2,600 / 1,278 / 1,822 at 4 K	§6.3
Booking	convention turning a decision's swing and noise into an error rate	A0 (released) … A3, B; all referred to the coded rail swing	§5.3
Ternary g128	weights in {−1, 0, +1}, one FP16 scale per 128	1.72 bits per weight	[3]
Q8_0	llama.cpp activation block: 32 int8 values + one FP16 scale	dot-product SNR 45.5 / 42.5 dB (Gaussian / Student-t)	[11]; §5.1

Term	Meaning	Value at the operating point	Where
PTQ1_0, PQ2_0	PrismML's own GGUF packings, not in mainline llama.cpp: dense trits; one trit per 2-bit slot	1.75 bits per weight, 5.95 GB; 7.21 GB (files on disk)	[3]
3:2 counter (full adder)	adds three equal-weight bits into sum and double-weight carry	15 cells, depth 7 (2 half adders + OR)	Part I §6.5; §4.1–4.2
Wallace tree	column reduction by layers of 3:2 counters [14, 15]	236 counters in 8 stages	§4.2
Block sum, lane	32-wide int8 × ternary sum per slot; the datapath computing it	$0 \le U \le 8{,}160$; 8,169–8,455 cells	§4.1–4.3
Yardstick, pass line	the Q8_0 activation error, assumed as the runtime's (§3.1); lane error power may not exceed it	42.5 dB; 6.3×10^{-9} per decision	§5.1–5.2
QFA-3	three-input interference decoded by three comparators: an analog full adder	1.3×10^{-5} at 4 K (REF)	§6.2–6.3

Table T1. Glossary. Each fabric term is defined at first use in the text; values are quoted at the operating point (1 THz, n = 10^{12} cm^{-2}, M/M_th = 0.7, F = 2, 69-GHz slot) with the temperature where it matters. "§" without a Part refers to this paper.

2.5 What Part III keeps, changes and considered instead

When Part III adopts a construct from Part I or II, it keeps it, changes it or replaces it with a choice the record does not contain (Table T2). Kept: the cell, its operating point and Part I's logic vocabulary, although on the released cell its NOT has no density route and its threshold AND does not exist (in-model; note 2026-10-16; Section 2.1); F = 2 and buffer trees; the half adder and its "two half adders and an OR" full adder as reference; the record's dissipation, noise-figure, noise-margin and classical-error laws, called read-only (the noise-margin law a design rule, Section 2.1); and Part II's Gaussian calculus (exact rules for loss, interference and phase-insensitive gain on coherent pulses) for every analog number. Five departures are substantive:

- **Slot rate.** 69 GHz gated, not Part I's 0.25-THz baseline — the record's in-model frontier for F = 2 logic, not a choice, and neither confirmed nor moved at the bench's ratio bias (notes 2026-07-31, 2026-08-01, 2026-08-02; two-clock alternative: Section 8.1(e)).
- **Timing.** Part I's DLY-cell balancing, generalized to a 92-cell-deep lane without latches or registers (Part I allows small fast registers), regenerates every waiting wire at every level (2,616 buffers, +51 %, as the listing places cells); stations every 10–13 cells, not 10–24; skew-tolerant alternatives open (Section 8.1(d)).
- **Decision error.** Part II gives one per-decision error (A0); Part I quotes none. This Part adds four compositions of the record's pieces (A1–A3, B) and a yardstick, the Q8_0 activation error assumed for the runtime (Section 3.1).
- **Analog unit.** Part II's two-input QMAC-1, on a core Part II describes as four-quadrant (weights of either sign, Part II §4.1), becomes a three-input unsigned full-adder tile, QFA-3, with the sign in digital select logic, relaunched cell outputs [E7] as inputs (relaunched from a rail-level '1' that no run of the record delivers; note 2026-10-08), both readings of the unanchored rail plane, and N_op = 2,600 instead of 400.
- **Temperature.** The digital lane keeps the no-cryogenics lock if DG1 lands at A0 at the swing its deciding cells receive in operation, reported beside the error (Section 5.5); the analog lane is a Part-II cold-class design, priced with its cooler, as is a cooled digital lane.

Model-side alternatives — a lower activation precision, or a model taught to tolerate the fabric's errors — leave the fabric unchanged and are priced in Section 8.2.

Construct	Parts I / II	Part III	Alternatives considered

Construct	Parts I / II	Part III	Alternatives considered
Slot rate	0.25-THz design baseline (Part I); ≈0.1 THz flushed (note 07-23)	69 GHz (band 63–74 GHz), the gated F = 2 frontier at 0.7 × the analytic threshold (note 07-31; neither confirmed nor moved at the bench's ratio bias, notes 08-01, 08-02); 14.5 ps	two-clock menu: 0.25-THz transport (double-buffered cell pairs, note 07-23) or a gated 91-GHz F = 1 wire (note 07-31) over slower logic, a design choice (§8.1(e))
Station spacing	"every 10–24 gates" (Part I §6.2)	10.0–13.3 cells by the released law; 24 unreachable (note 08-25)	none beyond 13 cells under the 20-dB link budget (§8.1(f))
Full adder	"two half adders and an OR" (Part I §6.5); 15 cells, depth 7 as composed here (§2.1)	kept as the reference idiom	threshold full adder, 5 cells, depth 2, for a cell that thresholds a three-way sum and takes a depleting input; in the amplitude encoding the released cell has no threshold on summed fields and no depleting input for a carrier-frequency signal (note 10-16; §8.1(a))
Path balancing	one DLY cell in the half adder	every waiting wire regenerated each level: 2,616 buffers as the listing places cells	4- and 14-cell skew tolerance (0.6–1.6 / 2.0–5.6 slots at 69 GHz, open; note 10-13): 581, 189 (§8.1(d))
Fan-out	F ≤ 2; buffer tiles for more (Part I §6.3)	kept; operand sharing by F = 2 trees [D4]	select in CMOS, nothing shared (§8.1(b))
Memory	"compute in plasmons, store as charge"; small fast registers allowed (Part I §6.4; RTD latches, external, §9.8)	no registers; nothing held on the fabric between slots (a departure: Part I allows small fast registers)	pumped bistable registers (not modelled as registers; in-model a CW $2f_0$ drain-pumped cell latches and holds a bit, but gated in the slot re-levels no flip, note 10-17) or recirculating loops (unmodelled) for waiting carries (§8.1(d)); on-fabric coherent accumulation unbuildable (note 07-16)
Decision error	classical bit-error rate (BER) of a rail-restored decision (Part II)	booking A0, plus compositions A1–A3 and B [D5], all referred to the coded rail swing	which applies is gate DG1, reported with the deciding cell's input-referred swing (§7; note 10-08)

Construct	Parts I / II	Part III	Alternatives considered
Yardstick	none	the Q8_0 activation error assumed for the runtime, 42.5 dB (new; §3.1)	error-aware fine-tuning to relax it (§8.2(b))
Analog unit	QMAC-1: 2 inputs, 2 comparators; sign asserted for the core (Part II §4.1)	QFA-3: 3 unsigned inputs, 3 comparators	phase-native trits; dual-rail ternary; $K \geq 4$ counters (§6.1–6.2)
Analog inputs	quiet comb launch of every input	relaunched cell outputs [E7] (new)	quiet launch as the PERM convention (§6.3)
Rail plane	rail anchored at no plane (note 07-17); no clamp at either plane in the released model (note 10-08)	both carried: P2 (REF) and P1 (CONS) [E4] (§6.3)	—
N_op	400, "launch-budget-capped" (undefined; note 08-25)	2,600 at P2; 1,278 at P1	400 as a sensitivity: 2.3×10^{-2} at 4 K
Offset c	coded $1/\sqrt{12}$, open (note 07-15)	$1/\sqrt{12}$ in REF and CONS	c = 0.02 in PERM
Cell power	1.267 kW/cm^2 stored at 353 K; kinetic 1.59–1.67 kW/cm^2 (closure; the tracked bias open in graphene's own fluid, note 10-15)	300 K by τ^{-3} scaling [D3]; 353 K closure-corrected	Joule model below 255 K, uncertified [E8]
Launch noise	20-dB link budget (Part I); 10 % coded jitter (Part II)	Part II's jitter at the first decision after launch, referenced to the rail swing [D8]	on Part II's 2-bit decode (4 K, N_op = 400): proportional with per-symbol variances (÷4) or rail-referenced (×59) against the coded additive jitter; ×239 span (note 08-25)
Temperature	≤ 80 °C locked (Part I); cold classes as a fork (Part II)	digital lane warm if DG1 lands at A0 at the swing its deciding cells receive (§5.5); analog lane cold, with its cooler	cooled digital lane (§6.5)

Table T2. What Part III keeps, changes and considered instead, construct by construct. Bracketed labels [D…], [E…] name this paper's in-model extensions (Table TA3); "§" without a Part refers to this paper.

3. The workload: Ternary Bonsai 2 27B

3.1 The model (external)

Everything in this subsection is external, from PrismML's model card, configuration, reports and runtime code [3–5, 26] (Table T3). The model is derived from Qwen3.8-27B [10] with the architecture unchanged: 27.36 B parameters, 24.35 B in a 64-block language backbone, 2.54 B in the embedding table and LM head (the final projection onto the vocabulary) and 0.46 B in a vision tower (the card; its three parts sum to 27.35 B, and the whitepaper [4] gives 0.47 B). Three of every four blocks use Gated DeltaNet linear attention [16], a recurrent layer with a fixed-size state; every fourth uses full attention with a key–value (KV) cache. The hidden width is 5,120, the feed-forward network (FFN) width 17,408, the vocabulary 248,320. PrismML has since released smaller members of the family, Ternary Bonsai 2

2B, 4B and 9B (15 October 2026 [27]); the 27B card and whitepaper were unchanged on re-check (24 October 2026), and only the 27B model is used here.

The weights are ternary g128 (−1, 0 or +1, one 16-bit floating-point, FP16, scale per 128 weights), stored in a rotated basis: a fixed blockwise Hadamard rotation is folded into the weights and applied to the activations at runtime, as QuaRot [17] does against activation outliers. The representation costs 1.72 bits per weight: $\log_2 3 \approx 1.585$ bits per trit, plus the FP16 scale amortized over 128 weights, plus the few tensors kept in higher precision. In GGUF it ships in PrismML's own packings, PTQ1_0 (dense trits at 1.75 bits per weight, 28 bytes per 128 weights; 5.95 GB) and PQ2_0 (2-bit slots, 7.21 GB); the whitepaper [4] quotes 1.76 bits per weight for the PTQ1_0 file. Neither packing is in mainline llama.cpp, which merged runtime support for the model's Hadamard rotation on 11 October 2026 and runs the model in its own Q2_0 type; support for PTQ1_0 there is an open pull request (external; [28]). The model keeps 98.2 % of its FP16 parent's average over 14 thinking-mode benchmarks, and decodes at 28.1 tokens per second (tok/s) on an Apple M5 Pro laptop (27.5-W GPU rail) and 129.9 tok/s on an RTX 5090 at 1.95 J per token. The whitepaper [4] and the release note [5] give slightly different throughput and quality figures (83.9 against 85.4 over a 20-benchmark suite; up to 143 tok/s on the RTX 5090; 46.8 tok/s on the M5 Max); none of them enters any computation here.

One assumption sets the lane's precision. The card states no rule for how its runtimes quantize activations; PrismML's own code settles it per backend (external; its llama.cpp fork at tag prism-b10787-73bfe6c [26] and the runtime bundled with the MLX pack [3], read on 24 October 2026; the build behind the card's measurements was not inspected). The fork's CUDA path quantizes activations to Q8_1 blocks, 32 int8 values with an FP16 scale and an FP16 block sum, for both packings; its CPU path pairs PTQ1_0 with Q8_0 and PQ2_0 with Q8_K, a block of 256 values with an FP32 scale (as mainline ggml pairs its own TQ ternary types [11]; its Q2_0 type pairs with Q8_0); its Metal path keeps F32 activations and the MLX runtime FP16, so the card's Apple figures, measured on the Metal path, involve no int8 activation step. The lane adopts the Q8_0 block of the llama.cpp family [11] as its activation format — 32 8-bit integer (int8) values with one FP16 scale — an assumption that matches the fork's CPU path for PTQ1_0, mainline's Q2_0 path and its open PTQ1_0 request [28], and shares its 32-value int8 geometry with the CUDA path's Q8_1 (external). The lane's cells are unchanged under any of these int8 formats; only the host's scale bookkeeping, and the yardstick of Section 5, would move. A ternary MAC is then int8 × {−1, 0, +1}, and a block of 32 products shares one activation scale: the unit mapped onto the fabric.

Item	Value	Source
Base model	Qwen3.8-27B, architecture unchanged	[3, 10]
Parameters	27.36 B = 24.35 B language backbone (64 blocks) + 2.54 B embedding/LM head + 0.46 B vision tower (27 blocks); the parts sum to 27.35 B, and [4] gives 0.47 B for the tower	[3]; [4]
Attention	hybrid, 3:1 — 48 Gated DeltaNet linear-attention blocks [16] and 16 full-attention blocks	[3], config
Widths	hidden 5,120; FFN 17,408 (SwiGLU); 24 query / 4 KV heads, head dimension 256	config
Linear-attention heads	16 key heads and 48 value heads of dimension 128 (q, k, v projection 2,048 + 2,048 + 6,144 = 10,240)	config
Vocabulary; context	248,320; 262 k tokens	[3], config
Weight format	ternary g128: {−1, 0, +1}, one FP16 scale per 128 weights; 1.72 bits per weight as a representation	[3]
Rotation	blockwise Hadamard (block 1,024, fixed ±1 signs) folded into the weights; transform applied to activations at runtime [17]	[3]
Higher precision	26.2 M parameters (0.0976 % of the language model, not of the 27.36 B total): linear-attention recurrent-state path and normalization weights	[3]
Packings	GGUF files on disk: PTQ1_0 (dense trits, 1.75 bits per weight) 5.95 GB, PQ2_0 (2-bit slots) 7.21 GB ([4] quotes 1.76 bits per weight and 5.93 GB for the PTQ1_0 file); MLX 2-bit 7.67 GB + 0.92 GB FP16 vision tower	[3]; [4]

Item	Value	Source
Quality	98.2 % of FP16: 84.78 vs 86.32 over 14 thinking-mode benchmarks; math 96.57 vs 97.06; coding 89.42 vs 89.07	[3]
Decode, batch 1	≈47 tok/s Apple M5 Max; 28.1 tok/s M5 Pro (27.5-W GPU rail); 129.9 tok/s RTX 5090 at 1.95 J per token (PQ2_0); H100 SXM 86.9 / 113.9 tok/s (PTQ1_0 / PQ2_0)	[3]
Activations	not stated by the card; in PrismML's llama.cpp fork the CUDA path quantizes to Q8_1 (32 int8 + FP16 scale + FP16 sum), the CPU path to Q8_0 for PTQ1_0 and to per-256 Q8_K for PQ2_0, while the Metal path keeps F32 and the bundled MLX runtime FP16 activations; this paper assumes Q8_0 (32 int8 + one FP16 scale); configuration dtype bfloat16	[26]; [3]; [11]; assumption

Table T3. Ternary Bonsai 2 27B, external facts from its model card, configuration, reports and runtime code [3–5, 10, 26]. Decode rates are the card's token-generation measurements (tg128, batch 1); the Apple M5 Max figure is from the card's earlier pre-rotation build. The whitepaper and release note differ slightly (Section 3.1); none of their figures enters a computation.

3.2 The per-token workload

From the external inputs [D6], digital_lane_bonsai.py §1 counts 25,598,361,600 ternary MACs per token (Table T4; shares computed from its rows): FFN blocks 67 %, Gated DeltaNet blocks 22 %, full attention 7 %, LM head 5 %. With the embedding table the model has 26,869,760,000 ternary weights, which checks exactly against the files (PQ2_0 spends 6 bytes more than PTQ1_0 per 128 weights). PTQ1_0 streams 5.669 GB per token (one embedding row is read, not the table). The Gated DeltaNet state (+314 MB, +5.5 %) and, at a 32 k context, the FP16 KV cache (+2.147 GB, +38 %) stay off the fabric. In the lane's units a token is 799,948,800 block sums and 199,987,200 weight-group scales.

Matrix	Shape (out × in)	Count per token	Ternary MACs per token
GDN attn_qkv	10,240 × 5,120	48	2,516,582,400
GDN attn_gate	6,144 × 5,120	48	1,509,949,440
GDN ssm_out	5,120 × 6,144	48	1,509,949,440
Attention q + gate	12,288 × 5,120	16	1,006,632,960
Attention k	1,024 × 5,120	16	83,886,080
Attention v	1,024 × 5,120	16	83,886,080
Attention o	5,120 × 6,144	16	503,316,480
FFN gate	17,408 × 5,120	64	5,704,253,440
FFN up	17,408 × 5,120	64	5,704,253,440
FFN down	5,120 × 17,408	64	5,704,253,440
LM head	248,320 × 5,120	1	1,271,398,400
Total			**25,598,361,600**
Ternary weights incl. embedding			26,869,760,000 (= (PQ2_0 − PTQ1_0 bytes)/6 × 128)
Weights streamed (PTQ1_0)			5.669 GB per token
GDN state read + write; FP16 KV at 32 k			+314 MB (+5.5 %); +2.147 GB (+38 %)
Block sums of 32; FP16 weight scales			799,948,800; 199,987,200

Table T4. The ternary workload of one decoded token (digital_lane_bonsai.py §1). GDN = Gated DeltaNet linear attention. Per-matrix counts multiply the listed shapes by the number of blocks; the total and the cross-check against the file sizes are printed by the listing.

3.3 The decisive property

At batch 1, tokens per second are bandwidth over bytes per token [6, 7]: 88, 176 and 529 tok/s at 0.5, 1 and 3 TB/s (digital_lane_bonsai.py §5); the HBM3 high-bandwidth memory of an H100-class GPU delivers 3.35 TB/s [18], and accelerators of 2025–26 deliver 8 TB/s from HBM3E (NVIDIA B300 [29], AMD MI355X [30]) and 19.2–23.3 TB/s from HBM4 (NVIDIA Rubin, 19.2 TB/s per GPU in its rack

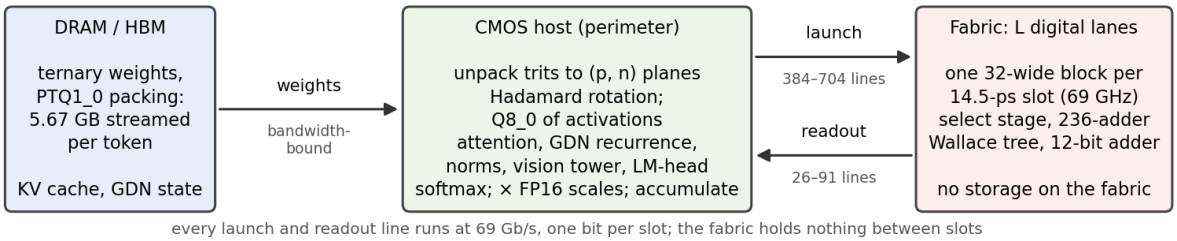
specification and up to 22 TB/s in its architecture description [31]; AMD MI455X, 23.3 TB/s [32]) (external). The 0.5–3 TB/s envelope of this paper therefore lies well below the 2025–26 flagships, at which the same rule gives 8 TB/s ÷ 5.669 GB ≈ 1,410 tok/s and 19.2–23.3 TB/s ÷ 5.669 GB ≈ 3,390–4,110 tok/s, bandwidths for which Section 4.6 computes no lane count. No arithmetic unit can raise those rates; the fabric can only keep pace cheaply, in area and energy, and Section 4 is written to that specification.

The card's measurements fit. The M5 Pro's 28.1 tok/s from the 7.21-GB PQ2_0 file is 28.1 × 7.21 GB ≈ 203 GB/s if the whole file is read, as the card's stated ≈204 GB/s evidently assumes (≈193 GB/s without the embedding table, as in Section 3.2). On the H100, 86.9 tok/s × 5.67 GB ≈ 0.49 TB/s is about 15 % of peak, which the card attributes to instruction throughput and launch overhead: unpacking trits costs arithmetic, which is why the host does it here (Section 4.1). digital_lane_bonsai.py §5 also prints three reported rates — RTX 4090 86.0 tok/s (PTQ1_0), a figure found in no primary source on re-check (the card gives 91.1 for PTQ1_0 and 81.2 for PQ2_0; external), and Section 3.1's press figures, 46.8 and 143 tok/s; none enters a computation.

4. The embedding: host, lanes and the digital lane

4.1 System partition

Figure T1 shows the partition. Weights stream from DRAM or HBM in the PTQ1_0 packing to a CMOS host at the fabric's perimeter, which unpacks each trit into two bit planes, p = [w = +1] and n = [w = −1]; applies the Hadamard transform (21.7–29.1 M additions per token) and Q8_0 quantization to the activations; and keeps every stateful or floating-point task — attention against the KV cache (0.81 G MACs per token at a 4 k context), the Gated DeltaNet recurrence (≈151 M FP32 MACs), norms, the vision tower and the LM-head softmax (digital_lane_bonsai.py §5). Each slot, a lane takes one 32-value activation block and the 32 matching weights of a row, and returns a 13-bit unsigned block sum U, which the host converts, scales and accumulates. Nothing is held on the fabric between slots.



What sets the speed (in-model, Section 4.6)

- tokens per second = memory bandwidth ÷ 5.67 GB: 88, 176, 529 tok/s at 0.5, 1, 3 TB/s
- one lane = 32 MACs per slot at 69 GHz = 86 tok/s of the 25.6 G MACs per token; 2, 3, 7 lanes keep pace
- the fabric adds area and energy, not speed: 17 k to 62 k cells, about 1.7 mm² at most, 0.13 to 0.64 W at 300 K
- the host scales 800 M block sums per token (1.27 TFLOP/s at 529 tok/s) and launches 27 to 49 Tb/s of operands

Figure T1. The system partition (Section 4.1). DRAM or HBM streams the PTQ1_0-packed ternary weights (5.67 GB per token; the KV cache and the Gated DeltaNet state also live there) over a bandwidth-bound link to a CMOS host at the fabric's perimeter. The host unpacks trits into (p, n) bit planes, applies the Hadamard rotation and Q8_0 quantization to the activations, runs attention, the GDN recurrence, norms, the vision tower and the LM-head softmax, and applies the FP16 group scales and accumulates. The fabric holds L digital lanes, each computing one 32-wide block per 14.5-ps slot (69 GHz) — select (p AND a) OR (n AND NOT a), a Wallace tree of 236 full adders and a 12-bit adder to a 13-bit block sum — with no storage on the fabric; 384–704 launch lines and 26–91 readout lines run at 69 Gb/s, one bit per slot. The text below the diagram states what sets the speed (Section 4.6): memory bandwidth ÷ 5.67 GB gives 88, 176 and 529 tok/s at 0.5, 1 and 3 TB/s; one lane is 32 MACs per slot at 69 GHz, 86 tok/s of the 25.6 G MACs per token, so 2, 3 and 7 lanes keep pace; the fabric adds area and energy, not speed (17–62 k cells, at most about 1.7 mm², 0.13–0.64 W at 300 K at the hydrodynamic tier); the host scales 800 M block sums per token (1.27 TFLOP/s at 529 tok/s) and launches 27–49 Tb/s of operands. In-model; nothing shown has been built.

No multiplier is needed because of a select identity. Each product bit is x_m = (p ∧ a_m) ∨ (n ∧ ¬a_m), m = 0 … 7: the product is a for w = +1, 0 for w = 0, and the one's complement ¬a = −a − 1 for w = −1, the missing +1 being folded into a per-block constant n_neg, the number of −1 weights in the block. Inverting each product's sign bit adds 128 and makes it an unsigned u ∈ [0, 255], so the block sum U = Σu satisfies $0 \le U \le 8{,}160 < 2^{13}$, and the host recovers the dot product with two constants:

$$u_i = [(p_i \wedge a_i) \vee (n_i \wedge \neg a_i)] \oplus 128, \quad 0 \le U = \Sigma_i\, u_i \le 8{,}160, \quad \Sigma_i\, w_i a_i = U + n_neg - 32{\cdot}128 \tag{T1}$$

The rest of the lane only adds bits, with the full adder, or 3:2 counter, which turns three bits of equal weight (inputs i_1, i_2 and a carry in c_in) into a sum bit s and a carry bit c_out of double weight while preserving their count:

$$s + 2{\cdot}c_out = i_1 + i_2 + c_in \tag{T2}$$

4.2 The lane netlist

The lane [D1] computes one block sum per slot (digital_lane_bonsai.py §2; Table T5; Figure T2). Its select stage builds (p ∧ a) ∨ (n ∧ ¬a) from four cells per product bit — two ANDs, a NOT and an OR — which is 1,024 cells for 32 products of 8 bits, and 32 NOT cells invert the sign bits. Each weight wire must reach eight select cells, which at F = 2 takes a six-cell buffer tree: 384 cells for the 64 wires. The 256 product bits then stand in eight columns of 32, one per bit weight.

A Wallace tree [14, 15] reduces the columns: each stage takes every column's bits three at a time into full adders, keeping sums in the column and moving carries one column up, until no column holds more than two bits. Eight columns of 32 take 236 full adders in 8 stages (per bit column: {0: 15, 1: 23, 2: 27, 3: 29, 4–7: 30 each, 8: 14, 9: 6, 10: 2}), and a 12-bit ripple adder yields the 13-bit result. At 15 cells per full adder the logic is 1,024 + 32 + 384 + 248 × 15 = 5,160 cells, all Part-I cells used as Part I uses them, with fan-out at most two.

Every inverting cell among them is a Part I Eq. (5) NOT that the released cell does not supply by any density route (in-model; note 2026-10-16): by note 2026-10-13's count of the depth model there are 783 — the select stage's 256, the 32 sign-bit NOTs and the 495 half-adder NOTs. In the amplitude encoding the lane uses, an inverter exists in-model only as destructive interference of the input with a coherent clock pulse (≥ 10 dB within 10.1–12.3° of phase and ±17.6–21 % of amplitude, by grid; $k \approx 2.9$ on Eq. (5)'s tanh scale; restoring at most marginally on the coarsest grid, open in the continuum) or behind a biased detector segment and an external electrical gain stage of ×9–12 that no Part contains; both are pump-fed by the clock burst, and the lane's ANDs, read as thresholds on summed fields, separate '1 + 1' from '1' by only 2.6–3.3 dB. Under phase encoding a NOT is a half-period delay, with restoration from a $2f_0$ pump admissible in-model only in a narrow window, and gated in the 69-GHz slot the pumped latch re-levels no flip (note 2026-10-17). A dual-rail lane, in which NOT is a wire swap, would need no inverter at the cost of carrying every signal with its complement and complementing every gate; its cell count has not been computed.

Two features are choices: the block width, Q8_0's 32 (Section 8.1(c)), and the record's composed full adder rather than a cheaper threshold one, which the record does not contain and which needs a threshold on summed fields that the released cell, in the amplitude encoding, does not supply (Section 8.1(a); note 2026-10-16).

Element	Cells	Note
Select stage	1,024	4 cells per product bit: (p ∧ a) ∨ (n ∧ ¬a), 32 × 8 bits
Sign-bit NOT	32	makes every product unsigned, eq. (T1)
Weight fan-out trees	384	64 wires × 6 cells, F = 2
Wallace tree	3,540	236 full adders × 15, 8 stages, 87 pass-through bit-stages
Ripple adder	180	12 full adders × 15, booked uniformly (the depth model builds the carry-free first bit as a 7-cell half adder: 8 cells, 0.1 % of the lane); 13-bit output
Logic	**5,160**	record idiom (2 half adders + OR = 15 cells per full adder)
Balancing buffers, tolerance 0 / 4 / 14 cells	2,616 / 581 / 189	+51 % / +11 % / +4 % on logic, as the listing places cells (as soon as possible, full-adder ports in list order)
Re-sync stations, cuts every 13 / 10 cells (the 13.3- / 10.0-cell law rounded down)	393 / 679	one per wire crossing each cut
Timing-closed lane, tolerance 0 (reference)	**8,169–8,455**	0.224–0.232 mm² at 27.4 µm² per cell
Timing-closed lane, tolerance 4 / 14 cells	6,134–6,420 / 5,742–6,028	if the gated cell tolerates a 0.6–1.6 / 2.0–5.6-slot skew (§4.3, §8.1(d))
Depth model	5,472 wires	critical path 92 cells (≈31 ps, 2.1 slots at Part I's ≈0.33 ps per cell, the listing's unit; in-model at the 69-GHz gated point, 13–37 slots in flight, up to 47 by the 90 % crossing; §4.3, note 10-13); longest wait 44 cells; outputs at depths 26–92

Table T5. The lane's cell budget (digital_lane_bonsai.py §2; in-model). The lane range spans the two station spacings, 13.3 cells (noise-figure floor; cuts every 13 cells) and 10.0 cells (Part I's NF ≤ 4 dB requirement). The lane's 783 inverting cells — the select stage's 256, the 32 sign-bit NOTs and the 495 half-adder NOTs — have no density route on the released cell (Section 4.2; note 2026-10-16).

4.3 Timing closure

On a wave-pipelined fabric every signal must reach its consumer in the slot in which it is needed. The depth model [D7] builds the netlist's 5,472 wires with one cell delay per gate (digital_lane_bonsai.py §2). The critical path is 92 cells: ≈31 ps or 2.1 slots at the ≈0.33 ps per cell of Part I's half adder at its 4-ps baseline (≈1 ps at depth 3, Part I Table 5; the listing's unit, digital_lane_bonsai.py §2), a unit that note 2026-10-13 contradicts in-model. At the 69-GHz gated point a cell's transient gain is set by drive duration (Part I §6.2), here a duty-0.8 gated 12-ps burst (note 2026-07-31). On the released solver one such isolated burst is delayed 2.7–4.0 ps per cell by its energy centroid and 2.0–7.3 ps across envelope estimators (demonstrated), and the cell's linear operator delays a narrowband envelope by 3.5–5.9 ps on the ratio bias leg (demonstrated per grid, in-model in the continuum); composed as 92 re-launched cells, the path holds 13–37 slots in flight at 69 GHz — up to 47 by the 90 % crossing and 33.8 by the energy centroid in a linear cascade, without the 7–9 stations it crosses, which would add 8–10 % if they sit in series (in-model; note 2026-10-13). The number of blocks in flight therefore stays open until gate DG2 measures the gated cell's delay; the cell counts do not depend on it. The 13 output bits leave at depths 26–92, each on its own readout line, and are deskewed at the receiver.

A carry produced early may wait up to 44 cells for its consumer. This lane has no registers (Section 2.1), a re-sync station re-times a pulse but cannot hold one, and a plasmon decays in about a picosecond, so the bit is regenerated at every level it waits, one buffer per cell delay, exactly as Part I's DLY cell does. At zero skew tolerance this costs 2,616 balancing buffers, +51 % on the logic, as the listing places cells — each as soon as its inputs allow, with full-adder ports filled in list order — which is not a minimum within the depth model (a netlist question, open). A 4- or 14-cell tolerance would need 581 or 189. In-model those skews are 0.6–1.6 and 2.0–5.6 slots at 69 GHz (2.0–5.75 ps per cell by the rise-to-centroid and shipped narrowband estimators, Section 8.1(d); to 2.0 and 7.1 by the 90 % crossing; note 2026-10-13), so a tolerant cell would have to accept an input between about half a slot and 1.6 slots late at a 4-cell tolerance, and two to nearly six slots late at 14. A gated burst's own delay does not depend on the preceding bit (to 0.01–0.5 ps; one cell, one pattern pair, the released grid) and falls by 0.7–1.1 ps from a small-signal drive to the top of the delivered band (2.6 ps by the 90 % crossing); the released gated statistic, a per-slot peak over the slot window, does not describe a late input, so the reference takes zero (Section 8.1(d)).

Stations are counted as one per wire crossing each cut of the depth axis: 679 at 10.0-cell spacing and 393 at 13.3 (cuts every 13 cells, the largest whole-cell spacing within the law; the earlier rule of one station per 10–24 logic cells gave 215–516). The timing-closed reference lane is 5,160 + 2,616 + (393 to 679) = 8,169–8,455 cells in the record-idiom booking, which charges every ripple bit as a 15-cell full adder (the depth model builds the carry-free first bit as a 7-cell half adder, 8 cells or 0.1 % fewer), 0.224–0.232 mm^2 at 27.4 µm^2 per cell (Table T5; Figure T2B). Its critical path crosses 7–9 stations, which sets the length of gate DG2 (Section 7).

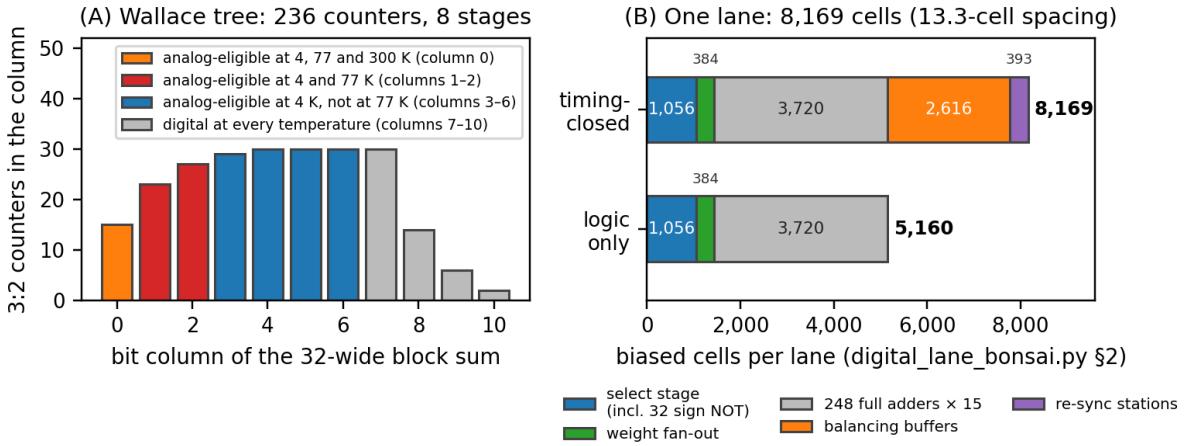


Figure T2. The lane (Sections 4.2–4.3 and 6.4). (A) The Wallace tree: 3:2 counters per bit column of the 32-wide block sum, 236 counters in 8 stages. Colours mark which columns' counters could be analog QFA-3 tiles at 41-dB dot-product SNR under the reference conventions (ternary_qmac_design.py §5, REF, Student-t activations), in three non-overlapping groups set by b∗, the highest bit column whose counters can be analog: column 0 at 4, 77 and 300 K (b∗ = 0 at 300 K); columns 1–2 at 4 and 77 K (b∗ = 2 at 77 K); columns 3–6 at 4 K but not at 77 K (b∗ = 6 at 4 K; 5 at 20 K and 3 at 48 K, Table T8); columns 7–10 are digital at every temperature. (B) One lane's cell budget (digital_lane_bonsai.py §2): logic only, 5,160 cells, against the timing-closed lane at the 13.3-cell station spacing (cuts every 13 cells), every segment labelled — select stage including the 32 sign-bit inverters (1,056), weight fan-out (384), 248 full adders (236 tree + 12 ripple) × 15 = 3,720, balancing buffers (2,616) and re-sync stations (393), 8,169 cells in all; at the 10.0-cell spacing the lane is 8,455 cells (0.224–0.232 mm^2 across the two). In-model.

4.4 Bit-exactness

The lane is exact by construction: every full adder preserves its count, eq. (T2), every product enters the tree as an unsigned u, eq. (T1), and $U \le 8{,}160 < 2^{13}$ cannot overflow 13 bits.

Demonstrated (runnable); a netlist check, not physics. digital_lane_bonsai.py §3 runs Bonsai-shaped matrices through the gate-level netlist with Boolean decisions — the full-attention k projection (163,840 block sums, 5.2 M MACs), the full Gated DeltaNet ssm_out projection (983,040; 31.5 M) and 1,024 rows of an FFN down projection (557,056; 17.8 M) — and every block sum is exact, the host's scaling reproducing the int8 reference bit for bit. The weights are synthetic and ternary, with an assumed one-third share of zeros (measured shares reach up to 51.5 % across 29 ternary models, external [33]; limitation 10); the activations are Gaussian and Q8_0-quantized, so the only gap to unquantized floating point is Q8_0's own 45.0–45.6 dB. This checks the netlist and the host's arithmetic; whether real cells decide correctly often enough is Section 5's question.

4.5 Power, energy, heat

At 300 K, with the hydrodynamic-tier τ^{-3}-scaled dissipation [D3] and no kinetic correction to it (at 353 K the closure of Part I §7.5 raises the dissipation ×1.25–1.32, Part I §8.2), a lane draws 64–86 mW, 29–39 fJ per MAC at full utilization; at the 353 K bias, closure-corrected, it draws 130–141 mW, 59–64 fJ per MAC (digital_lane_bonsai.py §2; in-model). Every correction the record evaluates would raise the 300 K figure: the closure raises the 300 K threshold by 7–16 % (note 2026-07-13), for which no dissipation is booked, and in that note's §6 single-rate finite-temperature kinetic operator a tracked 300 K bias exists only at the top of its electron–electron rate band with the drift heat shed, where the released Joule form gives ×3.17 the stored 353 K cell dissipation, while at 353 K, with the drift heat shed, none exists at the central and lower rates of the band and, at its top, none only under the note's projected treatment of the DC drive (in-model; note 2026-10-15; no lane figure computed). The 69-GHz slot was adjudicated at 353 K and is unmodelled at 300 K. At the released dissipation, heat is no constraint: at a 3.6 % fill the lane rises 2.8–3.7 K (300 K) or 5.8–6.1 K (353 K) against Part I's 55 K budget (digital_lane_bonsai.py §5); the configurations of Section 4.6 spend 0.9–3.3 mJ per token.

That is not where a decode's energy goes: reading 5.67 GB from HBM at ≈4 pJ per bit (HBM2-class, [19]) costs 5.67 GB × 8 bit/B × 4 pJ/bit ≈ 0.18 J per token, so the fabric adds about 0.5–2 %, and a GPU decode measures 1.95 J per token of board power (RTX 5090; external). Not booked are the CMOS perimeter and the wall-plug launch energy, which the record cannot fix because the launch-coupling efficiency is unquantified (Part I §9.5; the gated burst costs ×3.4–4.9 energy per '1' over the released 3-cycle burst, while time-averaged launch power rises only ×1.25–1.30 because its slots are 2.75–3.75× longer — demonstrated waveform integrals, in-model as a cost on the launch and clock subsystem; note 2026-07-31); no note prices the clock's distribution either — power per launch point, number of launch points, master-to-station loss (issue #139) — or the energy per bit of an electrical write and read (issue #141). The claim, if the cell works, is that the fabric's share of the arithmetic energy is small — not that the system is efficient.

4.6 The system

One lane does 32 MACs per slot, 2.208×10^{12} MACs per second at 69 GHz, or 86.26 tok/s of Bonsai's 25.6 G MACs per token (78.8–92.5 tok/s over 63–74 GHz). The memory sets the token rate, so the lane count follows from bandwidth (Table T6; Figure T5A; digital_lane_bonsai.py §5): 2 lanes at 0.5 TB/s (51 % busy), 3 at 1 TB/s (68 %), 7 at 3 TB/s (88 %). Lanes compute different rows against the same activation block, which is launched once and shared through F = 2 buffer trees of $256\cdot(2L-2)$ cells [D4]. Seven lanes take 60,255–62,257 cells, 1.65–1.71 mm^2 and 0.47–0.64 W at 300 K (0.96–1.04 W at 353 K).

The input–output count is the harder number: 64 weight and 13 readout lines per lane plus 256 shared activation lines — at 3 TB/s, 704 launch lines (48.6 Tb/s) and 91 readout lines (6.28 Tb/s) at 69 Gb/s each, 3.14 launched bits per MAC. The ≈800 lines — modulated, phase-locked and coupled into the acoustic plasmon at an efficiency the record does not quantify — are harder engineering than the lane.

The read crossing is not booked: the emulations flip logic decisions and launched bits only (Section 5.2), and only the launch side is priced (Section 5.4). Held to DG1's per-decision pass line, even a homodyne-limited decision would need ≥ 406 quanta at 300 K (477 at 353 K), more than the 99–277 a loaded F = 2 cascade sustains on the absolute and ratio bias legs (in-model; note 2026-10-08); a readout bit's own ceiling is not computed. Whether the lines could instead be electrical — written by a gate depletion and read by the cell's own rectification (note 2026-10-16; the gate response in-model, the rectification demonstrated, the gate's RC open) — is an open design question (issue #141).

The host's load is modest: scaling 800 M block sums per token at three operations each is 2.40 GFLOP (1.27 TFLOP/s at 529 tok/s), and the n_neg counts, fixed by the weights, can be precomputed and streamed (+10.6 % traffic) or counted during unpacking. Within 10^5 cells — the top of Part I's illustrative 10^4–10^5-cell scale (Part I §8.1), not a cap — 11 lanes give 949 tok/s, which a 3-TB/s memory feeds only with a batch of two or more.

Bandwidth	tok/s	Lanes (busy)	Cells; area	Fabric W at 300 K / 353 K	mJ per token	Launch lines (Tb/s)	Readout lines (Tb/s)	Launched bits per MAC
0.5 TB/s	88.2	2 (51 %)	16,850–17,422; 0.46–0.48 mm²	0.13–0.18 / 0.27–0.29	1.5–3.3	384 (26.5)	26 (1.79)	6.00
1.0 TB/s	176.4	3 (68 %)	25,531–26,389; 0.70–0.72 mm²	0.20–0.27 / 0.41–0.44	1.1–2.5	448 (30.9)	39 (2.69)	4.67
3.0 TB/s	529.2	7 (88 %)	60,255–62,257; 1.65–1.71 mm²	0.47–0.64 / 0.96–1.04	0.9–2.0	704 (48.6)	91 (6.28)	3.14

Table T6. The system at three weight-streaming bandwidths (digital_lane_bonsai.py §5; in-model plus arithmetic). Tokens per second are memory-bound (5.669 GB per token); one lane is 86.26 tok/s at 69 GHz. Cells include the F = 2 activation-sharing trees. Energy per token spans 300 K (low) to 353 K closure-corrected (high). Every line runs at 69 Gb/s.

5. The error budget: what "working" means

Section 4 shows that the lane is right if its cells decide correctly. This section asks how often they may not, against a yardstick taken from the runtime's assumed activation format (digital_lane_bonsai.py §4; Table T7; Figure T3).

5.1 The yardstick

This paper takes the error the runtime already accepts: Q8_0 quantization of the activations (the format assumed in Section 3.1) perturbs every dot product, so a lane adding no more error power at most doubles that perturbation. On PrismML's own runtimes (Section 3.1; external) the assumption holds for the CPU path with PTQ1_0; the CUDA path quantizes to Q8_1, of the same 32-value int8 geometry, the CPU path with PQ2_0 to the per-256 Q8_K, and the Metal path, on which the card's Apple figures were measured, keeps F32 activations, as the MLX runtime keeps FP16 — for those two the yardstick is an error the runtime does not incur. (The card's scores were measured with vLLM, so what Q8_0 itself costs in quality is not reported.) Over 4,000 synthetic dot products of length K = 5,120, Q8_0's signal-to-error ratio is 45.5 dB for Gaussian and 42.5 dB for Student-t ($\nu = 4$) activations; the reference uses 42.5 dB. This is a proxy: Bonsai's tolerance of lane errors is open, perhaps larger (it was trained against aggressive quantization) or smaller (lane errors are rare and large, not dense and small).

5.2 Error propagation and the pass line

A wrong decision in a high column changes the block sum by up to 2^{12}, one in column 0 by 1. digital_lane_bonsai.py §4 measures the average by gate-level emulation [D2], flipping every cell decision independently with probability BER (the error rate per decision). Over 163,840 blocks at BER = 1×10^{-5}, 2×10^{-5} and 4×10^{-5} the block-sum error variance per unit BER is $\kappa = 2.8\times10^8$ int² (13 % spread); flipping

launched bits instead gives 6.7–6.9×10^5. A K = 5,120 dot product is N_B = 160 blocks, so the lane's error power is N_B·κ·BER, and equating it to Q8_0's gives the pass line:

$$BER_ceil = P_sig \;/\; (N_B \cdot \kappa \cdot 10\text{^}(SNR_q/10)) \qquad \text{(T3)}$$

where P_sig is the dot-product signal power in the lane's integer units and SNR_q = 42.5 dB: 6.3×10^{-9} per cell decision and 2.6×10^{-6} per launched bit (in-model; emulation runnable), or 6.3×10^{-10} per decision for a 10-dB margin below Q8_0. The ceiling is an upper bound. The emulation flips only the 5,160 logic decisions of the netlist (8,361 flips in 163,840 blocks at 10^{-5}); the timing-closed lane adds 2,616 balancing buffers and 393–679 stations, which also decide and carry the weight of the wire they regenerate. Its κ is larger by an uncomputed factor, so 6.3×10^{-9} per decision, and every pass temperature below, is an upper bound (Section 9, limitation 6).

5.3 Five bookings of one decision

The record gives one error rate and implies several. Five bookings are carried, all assembled from released pieces [D5] (Table T7; Figure T3):

- **A0, released.** Part II's classical error of a rail-restored decision (qerrors.classical_ber: k = 8, full rail swing, matched noise of one restoring cell): 3.6×10^{-11} at 300 K (lane SNR 64.9 dB) and 9.1×10^{-10} at 353 K (50.9 dB). It passes below 394 K. Its margin is the parent chain's static noise margin, 0.2564 of the swing at k = 8 and 10-dB extinction, the same heuristic as Part I's noise margin, so the two Parts are not independent witnesses to it (note 2026-08-02; demonstrated, source read); k = 8 is the decision law's design rule, not a property of the released cell (Section 2.1; note 2026-10-16); and its swing is the coded rail, which no run of the record delivers to a deciding input (Section 5.5; note 2026-10-08).
- **A1, F = 2 softening.** The same law at k_eff = 8/√2, the softening note 2026-07-21 derives for a cell driven through an F = 2 split (as the lane's fan-out trees, select stage and half adders are), applied to every decision: 9.7×10^{-9} (40.6 dB) and 1.1×10^{-7} (30.1 dB). It passes below 292 K. A1–A3 call the same margin law at its default 10-dB extinction (digital_lane_bonsai.py §4; source read), the 10 dB that DG3's round floor specification also states, and k_eff inherits the design rule k_0 = 8 (note 2026-10-16).
- **A2, split.** A1 decided on the swing a driven cell receives after a −3 dB split: 3.6×10^{-5} (5.0 dB) and 1.2×10^{-4} (−0.4 dB). It passes below 144 K.
- **A3, junction.** A2 through one junction at the −1 dB optimistic end of Part I's −1 to −3 dB junction loss, the split-plus-junction charge of Part I's fan-out budget (Part I §6.3, §9.4; cf. the worst path of Part I Table 5, where a cell sits between split and junction): 2.0×10^{-4} (−2.5 dB) and 5.4×10^{-4} (−6.9 dB). It passes below 114 K; the −3 dB end is not computed and passes only colder.
- **B, Part II's calculus.** The Gaussian calculus with relaunched inputs [E4, E7], applied to a half adder read as a two-input interference decoded at the rail (each arm at 3,098 quanta, 0.81 of the rail, at which the summed top level reaches the rail at the combiner output after the input transit; 3,238 at 353 K): 5.5×10^{-3} per decode at 300 K (−4.0 dB by emulation), 1.0×10^{-2} at 353 K (−6.7 dB by emulation, −6.6 dB mapped) and 1.2×10^{-6} at 77 K (32.6 dB). It passes only below 59 K (digital_lane_bonsai.py §4: reference lane, 42.5-dB pass line). The 77 K SNR is mapped onto the per-decision axis through the 300 K emulation (0.051 equivalent cell errors per decode error), since the direct 77 K emulation prints 35.4 dB from only ≈25 error events; the analog listing's 32.4 dB (Section 6.6) agrees with the mapped value within 0.3 dB.

At 77 K every A booking is far inside the line (5.7×10^{-37} to 3.2×10^{-12}), while B, at 32.6 dB, is still outside. None of A1–A3 or B is a record value; each composes record pieces as the record does not, which is why they are bookings, not predictions. A0 is a record value, but of a swing the record assumes and never delivers to a deciding input (note 2026-10-08), so it too is not a prediction.

Booking	Assumes	300 K: error (SNR)	353 K: error (SNR)	77 K: error (SNR)	Passes below
A0	released classical BER: k = 8 (a design rule; margin 0.2564 at 10-dB extinction), full rail swing, matched noise of one restoring cell (Part II)	3.6×10^{-11} (64.9 dB)	9.1×10^{-10} (50.9 dB)	5.7×10^{-37} (322.9 dB)	394 K
A1	A0 at k_eff = $8/\sqrt{2}$, the F = 2 softening (note 07-21)	9.7×10^{-9} (40.6 dB)	1.1×10^{-7} (30.1 dB)	5.6×10^{-28} (233.0 dB)	292 K
A2	A1 on the swing after a −3 dB split	3.6×10^{-5} (5.0 dB)	1.2×10^{-4} (−0.4 dB)	6.3×10^{-15} (102.5 dB)	144 K
A3	A2 through one −1 dB junction (Part I §6.3's split-plus-junction charge)	2.0×10^{-4} (−2.5 dB)	5.4×10^{-4} (−6.9 dB)	3.2×10^{-12} (75.5 dB)	114 K
B	Part-II calculus [E4, E7] on a rail-referenced half-adder decode (per decode; each arm 3,098 quanta, the sum at the rail at P2)	5.5×10^{-3} (−4.0 dB)	1.0×10^{-2} (−6.7 dB)	1.2×10^{-6} (32.6 dB, mapped)	59 K
Pass line	lane error power = Q8_0's (Student-t); 10-dB margin: 6.3×10^{-10}	6.3×10^{-9} (42.5 dB)	6.3×10^{-9} (42.5 dB)	6.3×10^{-9} (42.5 dB)	—

Table T7. Five bookings of one cell decision (digital_lane_bonsai.py §4; in-model). Errors are per cell decision (per half-adder decode for B); the bracketed SNR is the lane's digital-error SNR for a K = 5,120 dot product with Student-t activations. Pass temperatures are the highest at which the booking meets the pass line (upper bounds for the timing-closed lane; Section 5.2). Row B's SNRs at 300 and 353 K are the direct emulation; mapped onto the per-decision axis through the 300 K emulation (0.051 equivalent cell errors per decode error, KAPPA_B_RATIO), they read −4.0 dB (identical by construction) and −6.6 dB. At 77 K, where the direct emulation sees only ≈25 error events and prints 35.4 dB, the table gives the mapped 32.6 dB, within 0.3 dB of the analog listing's 32.4 dB (Section 6.6). Every booking is referred to the coded rail swing (3,833 quanta): A0 and A1 decide on it, A2 and A3 on its split and junction fractions, and B caps its summed top level at it (3,098 quanta per arm at 300 K); no run of the record delivers that swing to a deciding input. At note 2026-08-01's loaded fixed point A_fix (≈75 quanta) and at the central F = 2 fixed point (133–201 quanta) all five fail at every temperature; the F = 2 band's top (249–277 quanta) passes A0 only below 14–19 K, and the delivered F = 1 level (628–955 quanta) only below 61–96 K (in-model; note 2026-10-08).

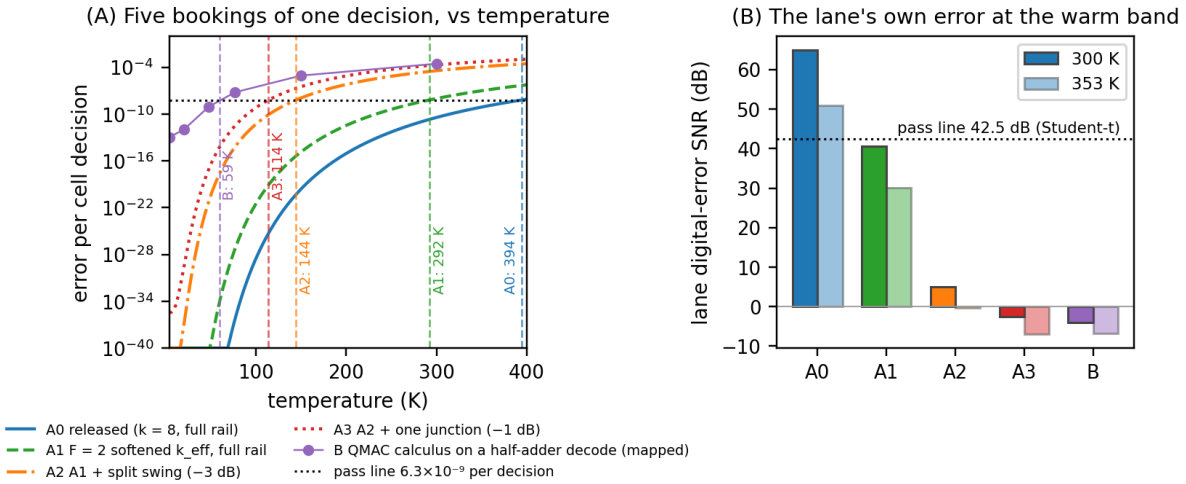


Figure T3. The error budget (Section 5; digital_lane_bonsai.py §4). (A) Error per cell decision versus temperature for five bookings of one decision, told apart by line style: A0 (released, k = 8, full rail; solid), A1 (F = 2-softened k_eff, full rail; dashed), A2 (A1 decided on the −3 dB split swing; dash-dot) and A3 (A2 plus one −1 dB junction; dotted), computed from the released chains; and B (the Part-II calculus applied to a half-adder decode; circles on a thin line) at 4, 20, 48, 77, 150 and 300 K, the per-decode errors of ternary_qmac_design.py §6 mapped onto the per-decision axis by the 300 K emulation's factor, 0.051 equivalent cell errors per decode error (KAPPA_B_RATIO, digital_lane_bonsai.py §4). The black dotted line is the pass line, 6.3×10^{-9} per decision (lane error power equal to Q8_0's, Student-t activations); the labelled dashed verticals mark the pass temperatures, A0 394 K, A1 292 K, A2 144 K, A3 114 K and B 59 K; the legend sits below the axes. (B) The lane's digital-error SNR at 300 K (solid) and 353 K (pale) for the five bookings against the pass line, 42.5 dB (Student-t); only A0 clears it, on the coded rail swing (note 2026-10-08). In-model.

5.4 Launch noise

Part II's coded launch — 20 dB, read as amplitude jitter of 10 % of the swing — added at the first decision after each launch [D8] makes each launched bit err 8.5×10^{-3} (k = 8) to 2.0×10^{-2} (k_eff) at 300 K, more than three orders of magnitude above the 2.6×10^{-6} ceiling. Meeting it needs a launch SNR of 27.9 dB (k = 8) or 30.9 dB (k_eff) at 300 K and 28.7 or 32.7 dB at 353 K — jitter at most 4.0, 2.8, 3.7 or 2.3 % of the swing (digital_lane_bonsai.py §4) — on all 384–704 lines: Part II's quiet-launch rule holds for a digital lane too. Note 2026-08-25 shows that the 20-dB figure stands for two different quantities in the released artifacts — a link budget against the thermal floor that Part I's spacing law consumes (Part I §6.2, §7.4) and the launcher's own amplitude repeatability (Part II's coding) — and leaves open which one Part I intends, at which plane. Its launcher-physics fork (the note's check P3, in-model), run on Part II's 2-bit decode at 4 K, N_op = 400 and 20 dB, spans ×239 in decode error across four launcher conventions — at the coded N_op plane a jitter proportional to each symbol's amplitude, booked with per-symbol variances, lowers the error about fourfold (either change alone leaves it unchanged), and a rail reference, itself anchored at neither plane, raises it up to 59-fold (open until the launcher's physics and reference plane are known). The figures above use Part II's additive convention referenced to the rail swing at the first decision after launch (digital_lane_bonsai.py §4 [D8]). They take the swing, and the jitter's reference, at the coded rail; at a smaller delivered swing — whatever the launcher supplies, whose level and plane the record leaves open — the margin shrinks at fixed relative jitter, so the errors would be larger and the launch requirement stricter (note 2026-10-08; direction only, not computed).

5.5 Reading the budget

Three readings follow, all in-model and all on the coded rail swing of 3,833 quanta. Only the released booking A0 passes warm, with 22 dB of margin at 300 K and 8 dB at 353 K. The record's own F = 2 softening, hard to avoid in a lane of F = 2 trees and half adders, puts the lane at the Q8_0 level: 40.6

against 42.5 dB at 300 K, a 1.9-dB fail, passing below 292 K. And if a decision's swing is what the driven cell receives after the split and junction losses of Part I's fan-out budget (Part I §6.3) — the budget Part I uses for gain — the pass temperature falls to 144 K for the split alone and to 114 K through a −1 dB junction (colder at −3 dB, not computed). Which reading holds is the open question of where the coded rail swing sits relative to the deciding cell's input (notes 2026-07-15, 2026-07-17). Whichever plane the coded rail belongs to, a decision compares its swing with input-referred noise, so the swing that counts is the one at the deciding cell's input — reading it at the intracavity plane would be a plane error — and no run in the record delivers a rail-scale swing there (in-model; note 2026-10-08). At the released booking of the decision noise the A0 law needs at least 2,924 quanta at the deciding input at 300 K (0.76 of the rail) and 3,434 at 353 K (1,654 and 1,925 at the quantum-limited (Caves) added-noise floor, a vacuum idler, of note 2026-08-05's idler law). At the swings a loaded F = 2 cascade sustains it reads 0.040–0.147 at 300 K (353-K swings with 300-K noise), and at note 2026-08-01's loaded fixed point A_fix (≈75 quanta) A0 reads 0.18, A1 0.22 and B 0.44 per decision or decode at 300 K, B 2.5×10^{-2} at 4 K, with no pass temperature: under the five bookings nothing the record delivers passes warm (in-model; note 2026-10-08, whose B reading generalizes the listing's construction to non-rail swings).

The lane, exact as a netlist, therefore works warm only if a loaded cell decides as well as the released booking says — one measurement, a half adder's decision error at fan-out 2, through real junctions, at rail drive, at 300 K and 353 K, with the input-referred swing at the deciding cell, its origin (fabric or bench) and the regime (re-launched or cascaded) reported beside it: gate DG1 (Section 7). If DG1 lands at A0 on a swing the fabric itself supplies — the A0 law needs a source-plane amplitude of at least 0.0874 of the density at 300 K and 0.0947 at 353 K at the released booking and calibration (in-model) — the lane is a room-temperature design; at A1 it wants about 8 K of cooling below 300 K or about 2 dB more model tolerance (61 K or 12.4 dB at the 353 K bias; Section 8.2); at A2–A3 it is a 114–144 K design, reopening Section 6.5; at B it is cryogenic — the A1, A2–A3 and B readings all on the coded rail swing (note 2026-10-08).

6. The analog alternative: the Part-II fabric as a counter tree

Part II's fabric adds by interference — pulses superpose passively at a junction and threshold cells of Part I's design (Section 2.1) decode the sum — so interference might replace some of the Wallace tree's 236 full adders; ternary_qmac_design.py asks whether (Tables T8, T9; Figure T4). Every number here carries two caveats: a second independent check of that listing raised five rework items that remain open (Appendix TA), and the relaunched '1' of REF and CONS is the coded rail level, which no run of the record delivers to a deciding input (limitation 15).

6.1 Why the phase-native trit is tempting, and why it loses

The tempting design puts the weight into the physics: +1 launches the activation's pulse at phase 0, −1 at phase π, 0 not at all, and interference forms the signed sum. Part II asserts sign-programmable weights only in text (Part II §4.1); no released call computes a signed tile. A signed tile built here [E6] with a coherent offset arm, summing two signed trits (five levels on four comparators), errs 5.4×10^{-4} at 4 K and 3.3×10^{-2} at 77 K, against 3.5×10^{-7} and 1.3×10^{-3} for the unsigned three-input counter below under the same quiet conventions (ternary_qmac_design.py §3): a signed sum crowds its levels around the offset, which also spends rail headroom.

Dual-rail balanced ternary — separate rails for +1 and −1 contributions, recombined by powers of three — fares worse: five trits span only ±121, so int8 needs a sixth trit or a requantization, and even lower-bound counts are 387–509 counters per block against 236 for binary columns (1.64–2.16×). The sign therefore stays arithmetic, in the select stage, and every analog input is an unsigned bit — a departure from the four-quadrant core Part II asserts for QMAC-1.

6.2 The K-input counter and the split law

An unsigned K-input counter interferes K pulses of N_op quanta in a balanced combiner tree [E1] and must separate K + 1 levels with K comparators. Each comparator needs the whole sum, so a K-way fan-out divides it (extension [E2] charges 3 dB per halving and $\lceil \log_2 K \rceil$ junction levels), and the per-comparator level spacing obeys a split law (ternary_qmac_design.py §2):

$$\Delta^2(K) = (2N_op/K) \cdot \eta_in \cdot \eta_extra \cdot \eta_fo(K) \to 2N_rail \cdot \eta_fo(K)/K^2 \propto K^{-3} \quad \text{(T4)}$$

where Δ is the level spacing in Part II's quadrature units (field amplitude, vacuum variance ½), N_rail ≈ 3,833 quanta is the rail, and η_in, η_extra, η_fo are the transmissions of the input transit, extra junction levels and comparator fan-out. The arrow caps the top level at the rail at the combiner output (plane P2), and the K^{-3} is rough, since η_fo carries the 1/K split; at P2, Δ^2 at 77 K with a quiet launch falls as 673, 140, 59 and 5.2 for K = 2, 3, 4 and 8. A counter must also remove bits — K in, $\lceil \log_2(K + 1) \rceil$ out — so K = 2 (QMAC-1, a half adder) removes none and K = 3 or 4 removes one. At K = 2 the tile reproduces Part II's qmac.error_2bit bit for bit at 48 points spanning temperature, N_op, junction loss, launch noise and propagation convention (demonstrated (runnable); ternary_qmac_design.py §1). Under REF (below), K = 3 errs 1.3×10^{-5}, 3.9×10^{-5}, 5.0×10^{-3} and 0.15 at 4, 20, 77 and 300 K, while K = 4 already errs 1.8×10^{-3} at 4 K and K ≥ 4 stays above 10^{-6} with every optimistic lever (2.4×10^{-6} under PERM). The tile to build is K = 3.

QFA-3 is that tile: three pulses interfere passively, a 1:3 split feeds three comparators, and the thermometer code $t_1t_2t_3$ (t_i = 1 if at least i inputs are 1) becomes sum = $(t_1 \wedge \neg t_2) \vee t_3$ and carry = t_2. In the record's idiom that is three comparators, a NOT, an AND, an OR and three path-balancing buffers — 9 biased cells — plus three optional carry-alignment buffers, 12 in all, which the listing charges against the full adder's 15. The comparators inherit Part II's k ≥ 16 redesign — a margin choice Part II makes for its three-level ladder (level gap x_range/2; Part II §5), carried here and into QFA-3's static-offset term without re-derivation for four levels — and it is not charged here. The decoder's NOT and AND are Part I's cells, which the released cell does not supply as Part I builds them (in-model; Section 2.1, note 2026-10-16).

6.3 Conventions and the operating envelope

Three conventions are carried, because the record leaves three things open (ternary_qmac_design.py §2, §4):

- **REF.** The rail caps the top level at the combiner output (P2, the first plane where the sum is one field). Inputs are relaunched [E7]: inside a tree each input is an earlier cell's output, so a '1' is a rail pulse (the coded rail level, which no run of the record delivers to a deciding input; note 2026-10-08, limitation 15) attenuated to N_op and a '0' a cell's unsaturated output carrying the released amplifier's noise, σ_0 = 3.56 % of the rail swing at 4 K. Extra junction levels add a quarter-wave each; c = $1/\sqrt{12}$ enters the decision's static-offset term [E3].
- **CONS.** REF with the rail capping the pre-loss launch sum (plane P1), so N_op = N_rail/3 = 1,278.
- **PERM.** A quiet comb launch, P2, multiport junctions, junction-only extra levels and c = 0.02.

Under REF, N_op = 2,600 quanta from 4 to 150 K — 6.5 times the registered 400 — set by a 1.69-dB relaunch attenuator from the rail, at 1.72 aJ per pulse (Table T8). The error stays ≤ 10^{-3} below 49.6 K (REF), 20.5 K (CONS) or 126.5 K (PERM), and ≤ 10^{-4} below 27.2 K, never, or 88.1 K; 10^{-5} and 10^{-6} are reached only under PERM, below 66.4 and 52.3 K.

From REF's 1.3×10^{-5} at 4 K, one change at a time: a quiet launch gives 3.5×10^{-7}, c = 0.02 gives 8.4×10^{-7}, multiport junctions 2.2×10^{-6}, the P1 plane 4.0×10^{-4}, the registered N_op (400) 2.3×10^{-2}, and −3 dB junctions 9.9×10^{-4}, flagged by the second check: in that row the P2 rail re-solve sets N_op = 6,530.5

quanta (ternary_qmac_design.py §4), 1.70 N_rail, so the relaunch attenuator of [E7], defined as a loss from the rail, acts as a +2.3-dB gain (an open rework item, Appendix TA). The rail plane alone moves the error 30-fold, and the record anchors it nowhere. A tile at 4 K stays within 2× of its quiet-launch error (3.5×10^{-7}) only if a '0' level's noise is $\sigma_0 \leq 1.56$ % of the rail swing, where the released amplifier rule gives 3.56 %, and the launch jitter is $\sigma_rel \leq 1.33$ % of the launched amplitude ($S \geq 37.5$ dB); at the coded 20 dB the error is pinned near 1.1×10^{-2}.

T (K)	REF: N_op; p	CONS: p	PERM: p	All-QFA tree SNR (REF)	b∗ (REF)	Analog counters REF / CONS / PERM
4	2,600; 1.3×10^{-5}	4.0×10^{-4}	4.1×10^{-13}	28.2 dB	6	184 / 94 / 236
20	2,600; 3.9×10^{-5}	9.6×10^{-4}	2.5×10^{-11}	23.4 dB	5	154 / 94 / 236
48	2,600; 8.8×10^{-4}	9.3×10^{-3}	4.1×10^{-7}	9.9 dB	3	94 / 38 / 236
77	2,600; 5.0×10^{-3}	3.0×10^{-2}	3.5×10^{-5}	2.4 dB	2	65 / 38 / 154
150	2,600; 3.3×10^{-2}	0.10	2.4×10^{-3}	−5.8 dB	1	38 / 15 / 65
300	3,338; 0.15	0.32	4.1×10^{-2}	−12.3 dB	0	15 / 0 / 15

Table T8. The QFA-3 tile and the hybrid lane (ternary_qmac_design.py §4–§5; in-model). p is the QFA-3 symbol error (not the weight bit plane p of Section 4); N_op is 1,278 throughout under CONS and 1,822 (2,065 at 300 K) under PERM. The SNR is that of a 5,120-long dot product (N_B = 160 blocks) with every counter analog (Student-t activations); b∗ is the highest bit column whose counters can be analog while the dot product keeps 41 dB, and the last column counts those counters out of 236.

6.4 The hybrid lane

The hybrid lane keeps the 236-counter tree of Section 4.2, makes the counters of columns 0 … b∗ QFA-3 tiles and leaves higher columns digital, since a wrong decode in column j costs 2^j. The functional tree is exact on 20,000 random blocks (demonstrated (runnable); a netlist check, not physics), and the error model matches Monte Carlo to 1.6 % (ternary_qmac_design.py §5). The analog listing uses its own Q8_0 estimator (42.7–44.6 dB) and a 41-dB target, 1.5 dB below Section 5's 42.5-dB pass line; the counts below are at that target, and an open rework item of the second check (Appendix TA) estimates its Student-t SNRs 1.7 dB optimistic from per-block scales (0.2 dB for Gaussian), a figure no listing prints.

An all-QFA tree reaches only 28.2, 23.4, 9.9, 2.4, −5.8 and −12.3 dB at 4, 20, 48, 77, 150 and 300 K (REF, Student-t), so the analog columns stop at b∗ = 6, 5, 3, 2, 1 and 0: 184, 154, 94, 65, 38 and 15 of 236 counters (Table T8; Figures T2A, T4B). CONS gives 94, 94, 38, 38, 15 and 0; PERM all 236 up to 48 K and 154 at 77 K. Column 0 qualifies at 300 K only because its counters carry weight 1.

The cell saving against an all-digital lane at the same temperature depends on accounting [E5] (QFA-3 at 12 cells against 15; four alignment cells per QFA in a mixed tree; full adders at 18 cells for wave-pipelined composition; the select stage in CMOS): −4 % to +25 % at 4 K, −3 % to +21 % at 20 K and −1 % to +9 % at 77 K (10.8 %, 9.0 % and 3.8 % in the record idiom: full adder 15 cells, QFA-3 12, select on fabric). The listing's all-digital reference (5,342–5,641 cells: its logic omits the 32 sign-bit inverters, 1,024 + 384 + 248 × 15 = 5,128 cells against Section 4.2's 5,160, and it adds a re-sync allowance at Part I's 10–24-gate rule, which Section 2.2 retires, with no wave-pipeline balancing) is smaller than the timing-closed lane, so only ratios are used, and the record-idiom savings are fractions of those 5,128 logic cells. Even if every counter were analog, the tiles would save at most about a third of the cells (≤ 32 %, with 18-cell full adders and the select stage in CMOS; 13.8 % in the record idiom), and only when cold.

6.5 Cooling economics

A cold lane pays for its cooler. The record's Joule model [E8], $P = n\, m_* v_0^2/\tau$ at the tracked bias (n here the carrier density), scales as τ^{-3} and makes a cold cell one-eighth of a 300 K cell at either end of its range (0.97 μW at and below 150 K, where the lifetime saturates, from the chain's stored 12.67-μW

value; the upper bound scales alike), and it is uncertified below ≈255 K, where the kinetic closure of Part I §7.5 has not been evaluated (Part I §8.3; Part II §8). Wherever graphene's own electron fluid has a tracked threshold — whether it has one at 353 K is open (Section 2.1) — its own Joule dissipation exceeds this form by a further enthalpy factor of 1.25–1.35 at the tracked drift, on top of what tracking that threshold costs (Section 2.2), and note 2026-10-15 evaluates graphene's own form only at 300 and 353 K (in-model). A cooler at efficiency η_c (a fraction of Carnot) spends (T_a − T)/(η_c·T) watts per watt removed — at Carnot 2.90 at 77 K, 14.0 at 20 K and 74.0 at 4 K from a 300 K ambient T_a — so a cold lane of N_cold cells draws, relative to the power P_warm of the warm all-digital lane of N_warm cells, a wall-plug power P_wall:

$$P_wall/P_warm = (N_cold/N_warm) \cdot r \cdot [1 + (T_a - T)/(\eta_c \cdot T)] \qquad \text{(T5)}$$

where r = P_cell(T)/P_cell(300 K) is the cold-to-warm ratio of cell dissipation. With r = 0.125 and an ideal cooler, the 77 K hybrid lane draws 0.47 and a 77 K all-digital lane 0.49 of the warm lane's power; 20 K and 4 K lanes draw 1.7 and 8.4 times and lose even at Carnot (Table T9; ternary_qmac_design.py §6). Break-even needs ≥ 39.6 % of Carnot for the 77 K hybrid and 41.4 % for the 77 K digital lane, and with equal cell dissipation (r = 1) every cold lane loses, by 3.8 to 67 times. Conversely, the hybrid breaks even at r ≤ 0.267 with an ideal cooler, 0.126 at 40 % and 0.035 at 10 % of Carnot; the Joule model's 0.125 sits on the 40 % column (Figure T4C).

The analog counters thus move the 77 K lane's power by about 4 % (0.47 against 0.49); the rest is the decision to cool, which the digital lane can make equally — and under bookings A2–A3 would have to. Whether a cooler of the needed efficiency exists at the needed size is a question for the cooler, not the fabric [20].

Lane	Analog counters	Cells ÷ warm lane	Equal dissipation, ideal cooler	Joule model (r = 0.125), ideal cooler	Carnot fraction needed	Break-even r at 100 / 40 / 30 / 20 / 10 % of Carnot
TQ-4 (4 K)	184	0.892	66.9×	8.37×	beyond Carnot	—
TQ-20 (20 K)	154	0.910	13.6×	1.71×	beyond Carnot	—
TQ-77 (77 K)	65	0.962	3.75×	0.468×	39.6 %	0.267 / 0.126 / 0.098 / 0.067 / 0.035
Digital at 77 K	0	1	3.90×	0.487×	41.4 %	0.257 / 0.121 / 0.094 / 0.065 / 0.033

Table T9. Cooling economics (ternary_qmac_design.py §6; in-model, cold-side dissipation uncertified). Wall-plug power is relative to the warm (300 K) all-digital lane, with Carnot work per watt removed of 2.90 (77 K), 14.0 (20 K) and 74.0 (4 K) from a 300 K ambient. The cell ratios are 1 minus the record-idiom savings of Section 6.4 (10.8 %, 9.0 % and 3.8 %). Break-even r is the largest cold-to-warm cell-dissipation ratio at which the cold lane matches the warm one. Every lane books its digital decisions at A0 (Section 6.6).

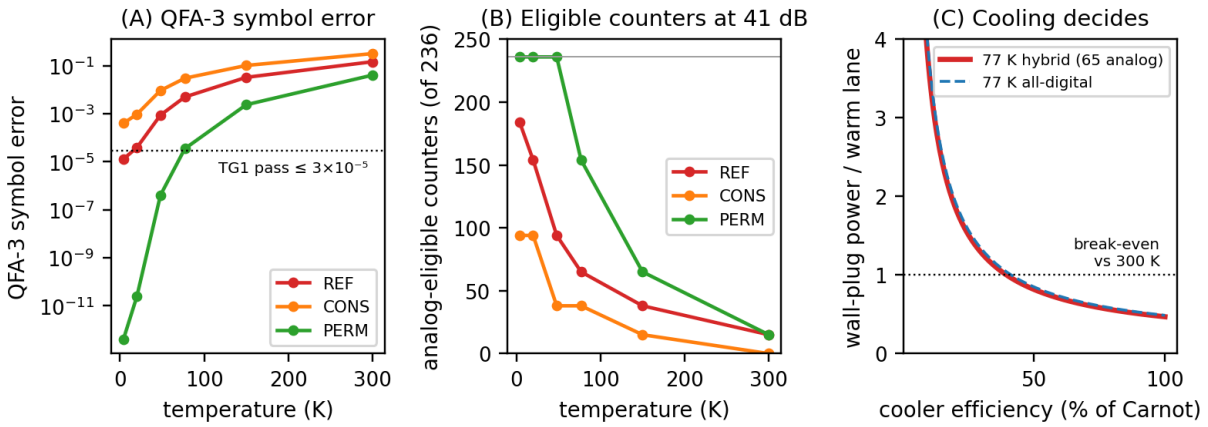


Figure T4. The analog counter tree on the Part-II fabric (Section 6; ternary_qmac_design.py §4–§6). (A) QFA-3 symbol error versus temperature under three conventions, marked by circles: REF (rail at the combiner output P2, relaunched inputs, c = 1/√12), CONS (REF with the rail at the pre-loss launch sum P1) and PERM (quiet launch, P2, multiport junctions, c = 0.02); the dotted line is gate TG1's pass threshold, 3×10^{-5}. (B) Counters of the 236-counter tree that can be QFA-3 tiles while the dot product keeps 41 dB, versus temperature, for the same conventions, with the same colours, markers and legend; the grey line is the full tree. (C) Wall-plug power of a 77 K lane divided by that of the warm (300 K) all-digital lane, versus cooler efficiency in percent of Carnot, under the record's Joule model (a cold cell dissipates 1/8 of a warm one): the hybrid lane (65 analog counters) solid, the all-digital lane dashed. The two coincide within a line width (0.468 against 0.487 at 100 % of Carnot) and cross the dotted break-even line near 40 % of Carnot. In-model; the cold-side cell dissipation is uncertified.

6.6 The booking fork

The hybrid lane of Section 6.4 books its digital columns at the released rate (booking A0 at 11 decisions per full adder: each half adder's five active cells and the OR; ternary_qmac_design.py §5 omits the four carry-alignment buffers that digital_lane_bonsai.py §4 also flips). Under booking B — the calculus that prices QFA-3, applied to the digital half adders — the analog listing's own all-digital lane (no balancing buffers) reaches 90.0 dB at 4 K, 50.8 dB at 48 K, 32.4 dB at 77 K and −4.2 dB at 300 K, meeting its 41-dB target only below 61 K (ternary_qmac_design.py §6); for the reference lane's logic netlist (likewise without balancing buffers; an upper bound, Section 5.2), digital_lane_bonsai.py §4 gives −4.0 dB at 300 K, 32.6 dB at 77 K (mapped; its direct 77 K emulation prints 35.4 dB from ≈25 error events) and a 59 K pass temperature against the 42.5-dB pass line (Table T7). The two listings agree within 0.3 dB (32.4 against 32.6 dB at 77 K), and the verdict does not move. The listing does not recompute the hybrid under B; under B the digital lane itself must run below about 60 K, so the comparison is again one between two cold lanes and their coolers (Section 6.5), its numbers not computed.

7. The decision, and the measurements that would change it

Keep the digital lane as the path for Bonsai: exact as a netlist (Section 4.4) and under 2 mm² for any memory up to 3 TB/s. Given a launcher that meets DG4 (which tests the launcher's jitter, not its launched level) and a clock source, itself an open design question (issue #139), it depends on one measurable quantity to work warm: DG1, with the deciding cell's input-referred swing reported beside it (note 2026-10-08). Keep QFA-3 as a test tile (gates TG1–TG5, beside Part II's QG family): the smallest analog counter that removes a bit, pre-registered under three conventions, testing relaunch noise, the rail plane and multi-input linearity. It is not a route to a better Bonsai lane: it saves at most about a third of the cells, only when cold, and the cooler then decides.

Both routes end at one measurement, the loaded decision error of a cell, which picks among bookings A0–A3 and B on one side and among REF, CONS and PERM on the other; the record cannot compute it

further, because the rail's reference plane relative to a deciding cell's input, the swing a loaded deciding cell actually receives (no run of the record delivers the coded rail to one; note 2026-10-08) and a loaded cell's effective sharpness (k = 8 is a design rule the released cell does not supply; note 2026-10-16) are open.

The gates (Table T10) are pre-registered in Part I's discipline and stand behind the prerequisites G1 (a cell has gain; note 2026-08-01, bench gate G1 reissued, proposes splitting it into a CW existence test, G1a, and a gated-drive knee report, G1b), L0 (Part I's linewidth data gate, Section 2.4; its protocol is posted as a measurement request, issue #88, and the staged bench plan, issue #97, runs it first) and QG1–QG3. DG1 is decisive: its threshold is the pass line, 6.3×10^{-9} per decision (6.3×10^{-10} with a 10-dB margin), and its targets are the five bookings, referred to the coded rail swing and spanning seven orders of magnitude in lane error power at 300 K, so a result names the machine only together with the swing at which it was obtained (below). DG2 tests what no earlier gate does — a balanced, re-synced cascade as deep as the critical path, at the slot rate, for which the record has no multi-cell transient and, in-model, a per-cell delay 6–22 times the ≈0.33-ps unit of Section 4.3 (note 2026-10-13); DG3 tests the gated slot, DG4 the launcher (8–13 dB quieter than coded, on hundreds of lines), DG5 the heat, and TG1–TG5 QFA-3 and the cold-cell assumption.

What the gates report beside a pass or fail follows from the record's qualifications; none moves a pass threshold. DG1 must report the input-referred swing at the deciding cell, whether the fabric or the bench supplied it, and whether that cell's '1' was re-launched or cascaded (DG1-S; note 2026-10-08): a bench-launched 10 % swing says nothing about a cell that receives the previous cell's output, and which regime this lane's datapath is in is open (note 2026-08-01, gated requirement round trip). DG1's half adder also presupposes a NOT and ANDs that the released cell does not supply as Part I builds them: its NOT has no density route, the in-model inverters being interference or an external gain stage, and it has no threshold AND, two '1's exceeding one by only 2.6–3.3 dB (in-model; note 2026-10-16; Section 4.2). Beside G1 and under its pins (bias 0.7 × the measured threshold, a carrier-frequency input at the delivered F = 2 level), that note registers two bench discriminants, shapes a measurement must match rather than gates (in-model): D-INV-1, the biased cell's DC drain photoresponse per input power is 4.4–5 times the unbiased one and the drain potential rises in both (a falling biased response, or a ratio below 2 or above 8, refutes the released rectification law); D-INV-2, the output against a top-gate density offset is a resonance curve, not a sigmoid, peaking +5 to +10 % above the operating density, at −0.4 dB per −1 % at the operating point (a sigmoid with a 10–90 % width below 0.10·n refutes it). Table T10 spells out what DG2's measured cell delay and L0's phase leg mean, after note 2026-10-13's discriminants PHX-1 … PHX-3.

If DG1 lands at A1, the warm lane needs about 8 K of cooling below 300 K or about 2 dB more model tolerance (61 K or 12.4 dB at the 353 K bias), priced in Sections 6.5 and 8.2. At A2 it is a 144 K machine and at A3 a 114 K one (colder at −3 dB junctions), under the arithmetic of Section 6.5, and at B a cryogenic one. All of these readings hold on the coded rail swing: under the five bookings nothing the record delivers passes warm, and at note 2026-08-01's loaded fixed point A_fix (≈75 quanta) and at the central F = 2 fixed point (133–201 quanta) A0, A1 and B pass at no temperature from 4 to 400 K (in-model; note 2026-10-08). If G1 fails, nothing physical in this paper survives.

Gate	Observable	Pass threshold	Model target (in-model) / source

Gate	Observable	Pass threshold	Model target (in-model) / source
G1	single-cell per-pulse net gain at ≤ 80 °C (prerequisite)	≥ +2.6 dB	+8 dB streaming, per-slot peak; +9.2 dB design-length rerun; drive pinned relative to the measured knee (Part I Table 7). Note 08-01 (G1 reissued) proposes, without amending G1, a split: G1a, the same bar on a retuned cavity with a CW small-signal drive ≥ 10× below the measured knee, an intracavity-peak observable and the bias at 0.7 × the measured threshold (+10.2 dB expected; the printed +8 dB is, in the note's reading, the shipped configuration's mis-tuned streaming value), and G1b, a knee and gated-drive report, not a pass bar (§2.2)
L0	sub-threshold cavity linewidth of a specular-contact cell, with contact-preparation contrast, temperature and drift sweeps (prerequisite)	near-model linewidth (Part I §10)	model half-linewidth 27.6 GHz; ≈3.2 GHz of excess half-linewidth per 1 % additive round-trip loss (Part I §7.5, §10; notes 07-18); the 27.6 GHz is the hydro-only anchor, ≈30 GHz with the closure's viscous supplement. Read on the phase leg it is PHX-1: a line-centre transfer-phase slope of −19.3 to −21.3 °/% of f_0 at G1's bias and G1a's small-signal drive (≥ 10× below the measured knee), referred to the cell's ports (each ps of uncalibrated path adds −3.6 °/%), against an in-model band of [−26, −15] °/% (note 10-13), not L0's threshold. In note 07-13 §6's single-rate kinetic bulk the electron–electron damping at 353 K is ×3.8 the closure's and the zero-drift line sits ≈12 % low (9.6–13.4 % across the rate band); once the contact channel is separated, L0's temperature sweep and joint width-and-shift reading also bear on the open current-channel rate (discriminants D1, D4, not falsifiers; in-model; note 10-15). Protocol: issue #88; run first in the staged plan of issue #97
QG1	two-pulse interference visibility at a junction combiner (prerequisite)	≥ 90 %	≥ 99 % (Part II Table Q4)
QG2	readout noise floor versus temperature, 300 → 4 K (prerequisite)	departs from the classical ∝ T line by ≥ 40 % at 20 K; flattens below ≈15 K	knee at T_Q ≈ 48 K (Part II Table Q4)
QG3	QMAC-1 2-bit symbol error at 77 K, N_op ≈ 400 (prerequisite)	≤ 10^{-2}	3.1×10^{-3} at −1 dB junctions with a quiet launch; 5.6×10^{-2} at −3 dB (Part II Table Q4); 1.9×10^{-2} at the coded 20-dB launch, additive at the N_op plane (Part II §4.2; note 08-25)

Gate	Observable	Pass threshold	Model target (in-model) / source
DG1	half-adder decision error, loaded (fan-out 2, real junctions, rail-level drive), at 300 K, 353 K and the lowest temperature reached; reported with the input-referred swing at the deciding cell, its origin (fabric or bench) and regime (re-launched or cascaded) (DG1-S, note 10-08)	$\leq 6.3\times10^{-9}$ per decision, an upper bound (§5.2); $\leq 6.3\times10^{-10}$ for a 10-dB margin	at 300 K: A0 3.6×10^{-11} · A1 9.7×10^{-9} · A2 3.6×10^{-5} · A3 2.0×10^{-4} per decision; B 5.5×10^{-3} per half-adder decode (mapped onto the per-decision axis in Figure T3A) (§5.3), all referred to the coded rail swing (3,833 quanta; A2 and A3 after the split and the junction; B at 3,098 quanta per arm under its rail cap); at note 08-01's loaded fixed point A_fix (≈75 quanta) A0 reads 0.18, A1 0.22 and B 0.44 at 300 K, and at note 10-08's central F = 2 fixed point (133–201 quanta) A0 reads 0.068–0.113, with no pass temperature for A0, A1 or B at either swing (in-model; note 10-08); the k = 8 behind A0 and A1 is the decision law's design rule, not a property of the released cell (note 10-16)
DG2	path-balanced, re-synced cascade ≥ 92 cells (7–9 stations) at the slot rate, with measured cell delay — at G1's bias and tuning, referred to the cell's ports: an isolated gated burst's energy-centroid lag and post-burst decay (PHX-2), the line-centre transfer-phase slope at G1a's small-signal drive (PHX-1, row L0), the AM-to-PM from 10× below to twice the measured knee (PHX-3) and the '1'-phase pattern spread (note 10-13)	open eye; per-decision error as DG1	the record has no multi-cell transient at the gated point: every gated row is a single cell re-launched each slot, the released chain has no cell-to-cell signal-transfer model, and whether a 92-cell datapath stays in that clocked regime or settles to the loaded fan-out fixed point is open (notes 08-01, gated requirement round trip; 10-08); per cell 2.7–4.0 ps by an isolated gated burst's energy centroid, 2.0–7.3 ps across estimators (released solver; demonstrated), so 13–37 slots in flight on 92 re-launched cells across the burst and narrowband estimators, up to 47 by the 90 % crossing (in-model; note 10-13) (§4.3)

Gate	Observable	Pass threshold	Model target (in-model) / source
DG3	gated F = 2 operation at the operating temperature, with note 07-31 §2's duty-0.8, slot-confined, data-gated launch at 0.7 × the device's measured threshold (note 08-01, G1 reissued; a proposal); reported with the '0'-slot floor at a named plane and in a named quantity, and with the amplitude and phase of one gated '0' slot after a '1' (a comb-referenced homodyne against the decoder's clock; note 08-12)	≥ 63 GHz with the '0'-slot floor ≥ 10 dB below data (report the strict 11.8-dB result too)	69 GHz (band 63–74 GHz) at 0.7 × the analytic threshold, 353 K (note 07-31). At the ratio bias (0.7 × the solver's grid threshold, also 353 K) the gain clears the measured requirement by +1.1 to +1.3 dB and the floor binds: under the round specification 67 GHz is robust and 71 GHz marginal; under the strict 11.8-dB anchor no rung is robust, 67 GHz sitting at the boundary (+0.007 dB at N = 240, a margin that flips under carrier phase and denomination, note 08-02) and failing by 0.36 and 0.50 dB at N = 480 and 720 (note 08-01, ratio-bias gain table); at its fitted convergence order 71 GHz is a 0.31-dB fail of the round specification and 67 GHz clears it at −10.88 dB (note 08-02, re-submitted; in-model). The model floor is the per-slot peak of the intracavity velocity perturbation: read in density it moves +0.055 dB at N = 240 and −0.445 dB at N = 480 at 67 GHz (note 08-02; demonstrated), and the slot-mode residue a decision consumes sits 3.83 dB lower (−15.9 to −15.7 dB below the mean '1' and 82–86° off its phase at 67 GHz, N = 240; 65–69° at N = 720; note 08-12; demonstrated at the model's ratio bias); no continuum or bench-referred phase is registered (note 08-13), so no phase target is set
DG4	launcher amplitude jitter σ_rel = σ_A/Ā of the launched '1', in the slot band at 69 GHz, over ≥ 10^4 slots (σ_rel = 10^(−S/20) in the released coding; note 08-25 §3), and line count (384–704 launch, 26–91 readout lines at 69 Gb/s)	launch SNR ≥ 28–33 dB (jitter ≤ 2.3–4.0 %)	record's coded launch: 20 dB (§5.4); the thresholds book σ_rel against the rail swing at the first decision after launch ([D8]), equal to note 08-25's post-coupling launch-plane σ_rel only if the launched '1' reaches that decision at the rail through a linear path, which the record never delivers (note 10-08)
DG5	lane temperature rise at 3.6 % fill	≤ 10 K	2.8–3.7 K (300 K); 5.8–6.1 K (353 K) (§4.5)
TG1	QFA-3 four-level symbol error, relaunched inputs, N_op at P2, 4 K	≤ 3×10^{-5}	1.3×10^{-5} REF; 3.5×10^{-7} quiet; 4.0×10^{-4} at P1; 9.9×10^{-4} at −3 dB junctions, a row whose rail re-solve sets N_op = 1.70 N_rail (an open rework item, Appendix TA; §6.3); every relaunched row (all but the quiet-launch entry) relaunches a rail pulse that no released cell emits, so each would move the worse way (not computed; note 10-08)
TG2	linearity of the 3-input combiner with its top level at N_rail	minimum level gap ≥ 0.9 × nominal	equally spaced by construction in the linear model
TG3	'0'-level output noise σ_0 and '1' jitter σ_rel (at the in-fabric, post-coupling N_op launch plane, in the slot band; note 08-25 §3) over ≥ 10^4 slots	σ_0 ≤ 1.56 % of the rail swing; σ_rel ≤ 1.33 % of the launched '1' amplitude (S ≥ 37.5 dB) — each the condition for p within 2× of the quiet-launch 3.5×10^{-7} at 4 K	released rule σ_0 = 3.56 % (fails; REF assumes it); coded launch 20 dB = 10 % (fails)

Gate	Observable	Pass threshold	Model target (in-model) / source
TG4	two-stage cascade QFA-3 → relaunch → QFA-3	error within [0.5, 2] × ($p_1 + p_2$)	independent errors add
TG5	cold-cell dissipation ratio r(77 K) at the tracked bias, and cooler efficiency η_c	r ≤ the break-even entry of Table T9 at the measured η_c	Joule model r = 0.125, uncertified (§6.5)

Table T10. Pre-registered gates. G1 and QG1–QG3 are prerequisites from Parts I and II; L0 is this Part's name for the linewidth test Part I §10 offers as a data gate that does not wait for the full kinetic solve, with no numeric threshold in Part I (note 2026-07-18, bench discriminants, pre-registers its bands). DG1–DG5 gate the digital lane and TG1–TG5 the QFA-3 test tile. DG1 is the decisive measurement. Pass thresholds are as pre-registered; model targets carry the record's qualifications, with notes cited by short date (note 10-08 = note 2026-10-08). DG4's thresholds book σ_rel as a fraction of the rail swing at the first decision ([D8]), TG3's at the in-fabric N_op launch plane; a pre-coupling bench RIN transfers to either only once the coupling side, the SNR denomination (quanta or quadrature, 2.65 dB apart at 300 K) and the slot-band integration are fixed (note 2026-08-25 §3, §5; open). TG1's threshold is among the open rework items of the analog listing (Appendix TA).

8. Alternatives and fine-tunings

This section asks what else Part III could have done with each construct it borrows (Section 2.5), and what the model side could do for the fabric. The numbers this Part computes here come from variants.py, extensions [V1]–[V5] (Tables T11, T12; Figure T5B), all in-model and none in the record; items (a) and (d)–(g) of Section 8.1 and item (a) of Section 8.2 also quote promoted notes, with their labels, and items 8.1(e), 8.2(a) and 8.2(b) external figures.

8.1 Fabric-side alternatives

(a) A native threshold full adder [V2]. The record's full adder is two half adders and an OR (15 cells, depth 7), yet the cell's primitive, in Part I's design, is a threshold on summed fields, and a full adder is two thresholds: the carry is the majority of the three inputs (one cell, all three enhancing, high threshold), and the sum is one cell with the three inputs enhancing and a weight-2 depleting coupling from the carry, firing when $i_1 + i_2 + c_in - 2 \cdot carry \geq 1$. With three alignment buffers that is 5 cells of depth 2, and lane logic falls from 5,160 to 2,680 cells. Part I §5 states the ingredients, but in the amplitude encoding the released cell supplies neither a threshold on summed fields nor a depleting input, in-model (note 2026-10-16): two in-phase '1's exceed one by only 2.6–3.3 dB on a transfer concave at every level, where a threshold needs it convex, and for carrier-frequency inputs the cell's only density coupling, rectification, accumulates, so a depleting gate input exists only behind a detector segment and an external electrical gain stage of ×9–12 (biased detector) that no Part contains. A three-input fan-in and a weight-2 coupling are, besides, not in the record, and the sharpness under a three-way sum is untested. [V2] is therefore a lever for a cell that thresholds, not for the released one: open, the cheapest fabric-side lever if such a cell exists, and (not computed) it would shorten the critical path behind the balancing cost. Under phase encoding a majority reads the sign of the summed field without a threshold (item (g)), but in-model it is admissible only at inputs ≤ ⅓ of the knee amplitude, where its weak class's ideal antipodal error is 0.15 at 353 K (note 2026-10-17).

(b) The select stage in CMOS. Moving the select to the host cuts the lane to 1,240 logic cells with the threshold full adder (on a cell that thresholds, item (a)), but every product bit must then be launched per lane — 8 bits per MAC with nothing shared — against 2 + 8/L (3.14 at seven lanes). That trades cheap cells for the most expensive resource, lines (Section 4.6). Considered and not adopted.

(c) Block width. Summing 64 or 128 products per slot (the weight group is 128) takes 491 or 1,002 counters, so logic per MAC stays near 161–164 cells, while readout bits per MAC fall from 0.406 to 0.219 and 0.117 and host scaling from 2.40 to 1.20 and 0.60 GFLOP per token (variants.py §1; extension [V1]). But a block wider than Q8_0's 32 sums activations with different scales, so it needs a runtime change — one activation scale per wider block — whose effect on the yardstick is not computed. Open.

(d) Timing tolerance. A 4-cell skew tolerance would cut balancing buffers from 2,616 to 581 (lane 6,134–6,420 cells), a 14-cell one to 189 (5,742–6,028; digital_lane_bonsai.py §2), if the gated cell accepts an input that late. In-model those skews are 0.6–1.6 and 2.0–5.6 slots at 69 GHz (to 2.0 and 7.1 by the 90 % crossing), at note 2026-10-13's per-cell delays (2.0–4.4 ps by a gated burst's rise-to-centroid estimators, 3.45–5.75 ps narrowband in the shipped configuration); the released gated statistic, a per-slot peak, does not describe such acceptance, and the saving stays open until DG2 measures the gated delay. A register alternative — Part I §6.4's small fast registers, holding a waiting carry instead of regenerating it — is not in the released chains and is not priced, nor are the RTD latches Part I §9.8 names as fast electronic registers (external devices). The nearest in-model candidate is note 2026-10-17's pumped cell: above a drain-current pump onset that lies above 0.70 and at most 0.72 of the DC current (N = 240; above 0.72 and at most 0.80 on N = 720, rising with refinement, continuum open), a $2f_0$ drain pump makes the released cell a latching parametric oscillator that holds a bit under a CW pump, its two states 180° apart and, at a 0.85 pump on N = 240, set by the sign of a seed; its latched level falls with refinement and, composed, would re-enter a next cell below the knee (≈0.003–0.008, N = 240). Gated in note 2026-07-31's slot at a 0.85 pump it amplifies a bit stream's flips ×1.75–3.15 over an unpumped cell (at $4–6\times10^{-3}$ inputs, ratio bias leg) but re-levels none: the level keeps the previous bit. Whether a contact can impose the needed 70–85 % current modulation at 2 THz, and at what energy, is open; no register is priced here.

(e) Batching and the two-clock menu. At batch 1 the lanes are 51–88 % busy; a batch of two or more fills them (Section 4.6). A two-clock architecture — 4-ps slots as an F = 1 transport and re-timing layer over slower logic — would rest on note 2026-07-23's double-buffered cell pairs (its option E4, at 2× area and energy; ROADMAP.md, a planning document of the repository, lists it under WP6 as an open design choice), and note 2026-07-31 finds that a duty-0.65 gated launch at 91 GHz survives as an F = 1 wire with no double buffering: +0.3–0.4 dB over the F = 1 CW corner at its one measured rung (N = 240; eye +15.1 dB), ≈ +0.9 dB after an in-model ladder transfer. The logic-to-transport ratio is then ≈1.3× (91 over 67–71 GHz), or up to ≈3.7× with 4-ps transport if E4 survives its open items: a design choice, not a result (in-model). Here one clock is kept: logic and launches run at the 69-GHz slot, each launch the slot-confined, data-gated duty-0.8 burst of the 1-THz carrier of Section 2.2 (note 2026-07-31; in-model), and nothing in the record prices a faster operand-transport layer beyond E4's 2× area and energy. Which source produces that burst is open: Part I's chain sets its repetition by its comb line spacing (Part I §3), no listing derives the 69-GHz burst from it, and the source class, the burst modulator and the distribution are the open design question of Section 2.2 (issue #139). External: a primary-source check of 24 October 2026 finds the best single resonant-tunnelling-diode oscillators near 1 THz at 15 μW (1.002 THz) and 73 μW (0.863 THz), operating temperature not stated [34]; the 0.73 mW reported at about 1 THz comes from an 89-element array, measured pulsed, whose elements are not intentionally coupled [35]; the best in-band photomixer it found, a uni-travelling-carrier photodiode, gives 24 μW at 0.914 THz [36]; and the check found no injection-locking or phase-noise data for resonant-tunnelling diodes at 0.8–1.2 THz and no chip-scale source of gated ≈1-THz bursts with picosecond edges at a tens-of-GHz rate.

(f) What cannot be done. Under Part I's 20-dB link budget against the thermal floor and a 6-dB margin, station spacing cannot exceed 13 cells. If the coded launch jitter is physical at the coded N_op = 400 plane, the same law gives 4.4–7.8 cells at 300 K (NF 2.8–4 dB, quanta or quadrature); booked at the rail it gives 1.0–1.7 cells at NF 2.8 dB, and in the self-consistent rail reading 12.2 — a ×12 spread set by a reference plane that is open (note 2026-08-25 §3, §5 item 1; a demonstrated function with an in-model input). A fixed lane cannot exploit weight sparsity, since every biased cell dissipates whatever its product. And nothing accumulates on the fabric across slots: the lossless coherent 16-slot combiner

behind Part II's published averaging column is unbuildable (note 2026-07-16), and the buildable version accumulates as charge off the fabric — as the host does here.

(g) Phase encoding with a $2f_0$ pump. Everything in this item is in-model on the released cell, whose solver fluid is Galilean, with no velocity cap (limitation 27); it gives consequences only, and nothing is priced here (note 2026-10-17). Under a 5 % source pump the cell restores a 0/π bit with a phase slope of 0.46–0.49 at knee-level swings, a window that narrows under refinement, to at or below the knee band's bottom in the continuum; at the delivered '1' restoration is weak (slope 0.76–0.87 under 1–2 % pumps). Restoration costs memory (7.8 ps against 5.5 ps unpumped, N = 240), so in note 2026-07-31's slot a flipped bit spends the slot overcoming the previous one, and even at knee-level swings an ideal antipodal read errs 1×10^{-4} to 7×10^{-3} at 353 K, not decision grade. Composed under note 2026-10-08's plane assumption, not run, a per-cell phase error settles to a fixed offset up to a 19.2° slip margin instead of accumulating. NOT is a half-period (0.5-ps) delay, ≈2.6 dB of passive line at 353 K; majority is read by the sign of the summed field without a threshold, but an in-domain majority input is at most ⅓ of the knee amplitude, ≈3.8 quanta, and its weak class's ideal antipodal error is 0.15 at 353 K. The scheme needs a $2f_0$ line phase-locked to the f_0 reference that AND and OR also need — two coherent lines instead of one (issue #139, open) — held within 20.2° uniformly or 11.0° from cell to cell, at ≈9× a delivered-'1' burst per slot by a mean-square proxy. Open.

Configuration	Counters (stages)	Adder / output bits	Logic cells: record · TFA · TFA + CMOS select	Launched bits per MAC, L = 2 / 3 / 7	Readout bits per MAC	Host scaling, GFLOP per token
b = 8, W = 32 (reference)	236 (8)	12 / 13	5,160 · 2,680 · 1,240	6.00 / 4.67 / 3.14	0.406	2.40
b = 6, W = 32	176 (8)	10 / 11	3,974 · 2,114 · 930	5.00 / 4.00 / 2.86	0.344	2.40
b = 4, W = 32	116 (8)	8 / 9	2,532 · 1,292 · 620	4.00 / 3.33 / 2.57	0.281	2.40
b = 2, W = 32	56 (8)	6 / 7	1,218 · 598 · 310	3.00 / 2.67 / 2.29	0.219	2.40
b = 8, W = 64	491 (10)	13 / 14	10,440 (163.1 per MAC) · — · —	6.00 / 4.67 / 3.14	0.219	1.20
b = 8, W = 128	1,002 (12)	14 / 15	21,000 (164.1 per MAC) · — · —	6.00 / 4.67 / 3.14	0.117	0.60

Table T11. The lane's size under the levers (variants.py §1; in-model, before timing closure). b = activation bits, W = block width, TFA = the threshold full adder [V2] (not in the record; it needs a cell that thresholds a three-way sum and takes a depleting input, while in the amplitude encoding the released cell, in-model, has no threshold on a two-input sum and no depleting input for a carrier-frequency signal, a three-way sum being untested; §8.1(a)). With the select stage in CMOS, every product bit is launched per lane: 8 bits per MAC at b = 8. Readout bits per MAC are output bits divided by W, and host scaling depends only on W (one scaling per block sum); for b < 8 both are computed from the listing's rows, not printed by it. Launched bits per MAC (2 + b/L) do not depend on W.

8.2 Model-side fine-tunings

The model can also move toward the fabric, using the techniques that made it ternary. None has been tried on Bonsai; the numbers below are in-model compositions on synthetic activations unless labelled otherwise, and whether the model keeps its quality under any of them is external and open.

(a) Activation precision. Quantization-aware fine-tuning lowers activation bits — BitNet a4.8 [21] reports 4-bit activations for 1.58-bit models at near-parity; LLM-QAT [22] is the general quantization-aware method, and SmoothQuant [23] a training-free post-training alternative. External: BitNet v2, trained with native 4-bit activations, again reports near parity, averaging 58.30 at 7B against 58.12 with 8-bit activations [37], whereas post-training quantization of full-precision 7–8B models to ternary weights with mixed-precision activations in a 4-bit setting (2–8 bits per layer) loses 11–17 points of average zero-shot accuracy against FP16, about 5–8 of them the activations' share against the same

ternary weights at 16-bit activations [38]. At b = 6, 4 and 2 bits the tree has 176, 116 and 56 counters, lane logic 3,974, 2,532 and 1,218 cells, and seven lanes launch 2.86, 2.57 and 2.29 bits per MAC instead of 3.14 (Table T11). The larger effect is on the error budget, since the model's quantization error grows faster than the lane's error weights shrink: the yardstick [V3] falls to 30.4, 17.9 and 3.5 dB (Student-t), κ_b [V4] to 1.8×10^{7}, 1.3×10^{6} and 6.8×10^{4}, and the ceiling rises to 9.5×10^{-8}, 1.3×10^{-6} and 1.3×10^{-5} per decision (variants.py §2). At 4 bits A1 passes throughout the scanned range (to 400 K, the scan top), A2 below 213 K and A3 below 168 K, against 291, 144 and 114 K at 8 bits in the same emulation (Table T12; 292 K with the reference lane's κ, Section 5), all, like Section 5's, on the coded rail swing (note 2026-10-08).

(b) Error-aware fine-tuning. Analog accelerators train with injected noise so that networks tolerate their hardware's errors [24]; at language-model scale (external), analog foundation models trained with noise injection stay comparable to 4-bit-weight, 8-bit-activation baselines under analog noise (Phi-3-mini, Llama-3.2-1B) [39], low-rank adapters lift Llama-3.1-8B's GSM8K accuracy under 6.7 % analog weight noise from 37.98 % to 70.74 % (85.06 % digital) [40], and on simulated faulty digital hardware error resilience grows with model size [41]. If Bonsai tolerated fabric error power 10 or 100 times its Q8_0 error [V5], the pass line would fall from 42.5 to 32.5 or 22.5 dB, and the pass temperatures would become 338 K or the whole scanned range to 400 K (A1), 168 or 200 K (A2), and 132 or 158 K (A3) (variants.py §3). The lane is unchanged; and since its errors are rare and large, the training noise would have to be the lane's own, not a Gaussian.

(c) Regrouping the weight scale. One FP16 scale per 32 weights instead of per 128 adds $16/32 - 16/128 = 0.375$ bits per weight to PTQ1_0's 1.75 bits per weight [3] — about +21 % weight traffic, and so about 18 % fewer tokens per second in a short-context, memory-bound decode (external arithmetic; no listing computes it). It buys nothing: each 32-wide block already lies inside one 128-weight group, so every block sum already takes one weight scale and one activation scale. Considered and rejected.

(d) Ternary activations. At b = 2 the lane is 1,218 cells and 56 counters, but the yardstick collapses to 3.5 dB: a different model, not a fine-tuning. Listed for completeness.

In sum, Part III keeps the record's full adder and select-on-fabric design, with its own generalization of Part I's DLY balancing, as its reference, and considers six fabric-side alternatives (noting three impossibilities) and four model-side ones: the threshold full adder and the skew tolerance are worth measuring — the first only on a cell that thresholds a summed field and takes a depleting input, which the released cell, in-model, does not supply in the amplitude encoding (two in-phase inputs exceed one by only 2.6–3.3 dB; note 2026-10-16), the second only if the gated cell accepts an input 0.6–5.6 slots late (note 2026-10-13); phase encoding with a $2f_0$ pump restores phase in-model only in a narrow window and is unpriced (note 2026-10-17); wider blocks wait on a runtime change; select-in-CMOS and the regrouped scale are rejected; and 4-bit activations and error-aware fine-tuning (in-model, on synthetic activations and the coded rail swing; whether the model keeps its quality under them is external and open, and external work reports near parity at 4 bits with quantization-aware training [37] but not for post-training quantization to ternary weights [38]) are the levers that move the pass temperatures.

Quantity	b = 8	b = 6	b = 4	b = 2	b = 8, 10× tolerance	b = 8, 100× tolerance
Pass line (yardstick, Student-t)	42.5 dB	30.4 dB	17.9 dB	3.5 dB	32.5 dB	22.5 dB
Yardstick, Gaussian	45.5 dB	33.1 dB	20.3 dB	3.6 dB	—	—
κ_b (int^2 per block per unit BER)	2.93×10^{8}	1.80×10^{7}	1.26×10^{6}	6.83×10^{4}	2.93×10^{8}	2.93×10^{8}
Ceiling per decision (Student-t)	6.1×10^{-9}	9.5×10^{-8}	1.3×10^{-6}	1.3×10^{-5}	10× the b = 8 value	100× the b = 8 value
A0 pass temperature	394 K	whole scan (to 400 K)	whole scan (to 400 K)	whole scan (to 400 K)	whole scan (to 400 K)	whole scan (to 400 K)
A1 pass temperature	291 K	350 K	whole scan (to 400 K)	whole scan (to 400 K)	338 K	whole scan (to 400 K)
A2 pass temperature	144 K	173 K	213 K	267 K	168 K	200 K
A3 pass temperature	114 K	137 K	168 K	211 K	132 K	158 K

Table T12. The error budget under the model-side levers (variants.py §2–§3; in-model compositions on synthetic activations). At b = 8, variants.py's emulation (1,024 × 5,120 rows at BER 2×10^{-5}) gives $\kappa = 2.93\times10^{8}$ and a 6.1×10^{-9} ceiling, within the 13 % spread of the reference lane's 2.8×10^{8} and 6.3×10^{-9}. The tolerance columns assume the model tolerates fabric error power 10× or 100× its Q8_0 error, a fine-tuning target rather than a record quantity. "Whole scan" means the booking passes at every scanned temperature, up to the 400 K top of the scan (variants.py prints "above 400 K"). Every pass temperature is read on the coded rail swing, as in Table T7; at the swings the record's loaded cascade delivers, no booking passes warm at b = 8 (note 2026-10-08), and the levers are not evaluated there.

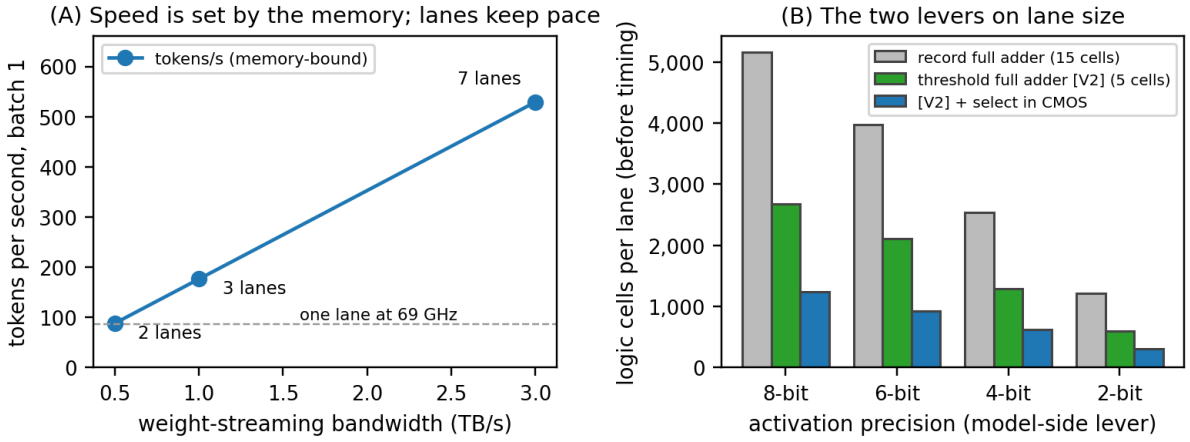


Figure T5. System scale and the levers (Sections 4.6 and 8). (A) Speed is set by the memory; lanes keep pace: tokens per second at batch 1 versus weight-streaming bandwidth, 88, 176 and 529 tok/s at 0.5, 1 and 3 TB/s (memory-bound, 5.669 GB per token), each point labelled beside it with the lanes needed to keep pace (2, 3 and 7); the dashed line is one lane at 69 GHz, 86.26 tok/s. (B) The two levers on lane size: logic cells per lane, before timing closure, versus activation precision (8, 6, 4 and 2 bits), for the record full adder (15 cells), the threshold full adder [V2] (5 cells) and [V2] with the select stage moved to CMOS — 5,160 / 2,680 / 1,240 cells at 8 bits down to 1,218 / 598 / 310 at 2 bits (variants.py §1). In-model; the threshold full adder is not in the record, and in the amplitude encoding the released cell supplies, in-model, neither its threshold on a summed field nor its depleting coupling (§8.1(a)).

9. What is demonstrated, in-model, external and open; limitations

- **Demonstrated (runnable); netlist and arithmetic checks, not physics.** The lane netlist is exact on 1.70 M Bonsai-shaped block sums with bit-exact host scaling; the analog tree is exact on 20,000 random blocks; the K = 2 counter tile reproduces Part II's qmac.error_2bit bit for bit at 48 points; the workload count matches the file sizes; the record anchors are released values. Rows the cited notes label demonstrated — runs of the released, unedited solver, such as a gated burst's delay through

one cell (note 2026-10-13) — are quoted with that label and remain statements about the released model (note 2026-10-08).

- **In-model.** Everything physical: timing-closed cell counts, power, energy, heat, bookings, pass temperatures, the launch requirement, QFA-3's envelope, eligible counters, cooling, every lever of Section 8 and the cited notes' compositions — the swings a loaded cascade delivers, the lane's latency, the threshold analysis in graphene's own electron fluid, the released cell's missing threshold AND and density-route NOT, and the pumped cell (notes 2026-10-08, 2026-10-13, 2026-10-15, 2026-10-16, 2026-10-17).
- **External.** Bonsai's architecture, sizes, scores and decode rates [3–5, 10], PrismML's runtime activation formats and later releases [26–28]; HBM bandwidth [18, 29–32] and access energy [19]; the Q8_0 format [11], whose use for Bonsai is this paper's assumption (Section 3.1); and the device and inference literature re-checked against primary sources on 24–25 October 2026: measured ternary zero shares [33], compact terahertz sources [34–36], 4-bit and error-aware inference [37–41] and scalable graphene [42, 43].
- **Open.** The cell's gain (G1) and the linewidth gate (L0) with the ±1 dB contact allowance; whether the cell has a DS threshold at 353 K in graphene's own electron fluid, which turns on the rate and the finite-drift structure of an open electron–electron relaxation channel, that of the current not locked to momentum (none in note 2026-07-13's single-rate kinetic operator at the central and lower rates of its band when the electrons shed their drift heat; three admissible two-rate continuations, identical at zero drift, give three answers; note 2026-10-15); the loaded decision error (DG1), the input-referred swing a deciding cell receives (note 2026-10-08) and so the booking; the reference plane of the rail, and so where the coded rail swing sits relative to a deciding cell's input; the lane's inverter route (interference or an external gain stage; in-model the released cell has no density route to a NOT and no threshold AND, note 2026-10-16) and the uncounted dual-rail alternative; the gated cascade and slot (DG2, DG3), including whether a 92-cell datapath stays in the clocked regime of the gated rows or settles to the loaded fan-out fixed point (notes 2026-08-01, 2026-10-08), the per-cell delay (note 2026-10-13), and the floor specification, floor observable and bias convention behind 69 GHz; the launcher (DG4); the clock source and its distribution (issue #139; Section 8.1(e)); whether a scalable graphene/hBN stack reaches Part I's best-case lifetime row (issue #140; limitation 1); an electrical boundary — the write, the read at the delivered swing (note 2026-10-08) and the electrical gain stage a gate-driven inverter needs (note 2026-10-16) — tracked as issue #141; which activation format the lane would serve (Q8_0 assumed; PrismML's own runtimes differ by backend, Section 3.1); the model's tolerance of lane errors and of lower activation precision; QFA-3 and its noise (TG1–TG4); cold-cell dissipation (TG5).

The limitations, merged across the three design listings (digital_lane_bonsai.py, ternary_qmac_design.py and variants.py), with five that the promoted notes add (limitations 27–31):

1. Everything physical is in-model and inherits an unproven gain cell; no cell, cascade, launcher or readout exists. The lifetime row beneath every gain and power figure (τ = 1.0 ps at 300 K) is Part I's best-case assumption, in premium-material territory, against a measured room-temperature baseline for propagating acoustic terahertz plasmons of $\tau \approx 0.4$–0.6 ps (Part I §4); whether a scalable stack reaches it is an open design question (issue #140). External: a primary-source check of 24 October 2026 found no 2025–26 experiment claiming net terahertz gain, emission or DS self-oscillation from drifting electrons in graphene or another two-dimensional channel, no 2024–26 graphene/hBN stack, scalable throughout, that beats the stack of Fukamachi et al. [42], and, in scalable (CVD) graphene, a measured terahertz plasmon $Q \approx 2$ at ≈2.5 THz in one device, with the resonances measured only cryogenically and vanishing above ≈130 K [43].
2. The lane netlist and depth model are this paper's [D1, D7]; the depth model counts one cell delay everywhere, with no spread. The listing times that unit at the ≈0.33 ps per cell of Part I's half adder

at its 4-ps baseline, a value note 2026-10-13 contradicts in-model: on the released solver an isolated duty-0.8, 12-ps burst is delayed 2.7–4.0 ps per cell by its energy centroid and 2.0–7.3 ps across envelope estimators (demonstrated, one re-launched cell), and a narrowband envelope 3.5–5.9 ps (in-model). A burst's own delay does not depend on the preceding bit (to 0.01 ps at small signal and 0.04–0.5 ps at a 0.0197 drive; one pattern pair, N = 240), but it falls by 0.7–1.1 ps (2.6 ps by the 90 % crossing) from a small-signal drive to the top of the delivered band, 0.027 (N = 240), and the carrier phase of a gated '1' spreads 4–12° peak to peak in two preceding-bit classes (demonstrated).

3. There is no layout or routing: areas are cell counts × 27.4 μm^2, without wires, guard cells, clock taps or line fan-in.
4. Bookings A1–A3 and B compose record pieces and are not record values; which applies depends on the open question of where the coded rail swing sits relative to the deciding cell's input (notes 2026-07-15, 2026-07-17), the swing that counts being the input-referred one. All five, A0 included, are referred to the coded rail swing or fixed fractions of it, none of which any run of the record delivers to a deciding input (note 2026-10-08), and the k = 8 behind A0 and A1 is the decision law's design rule, not a property of the released cell (note 2026-10-16).
5. Decisions are modelled as independent; correlated errors, disorder, drift and metastability are not (a metastability sensitivity is carried for QFA-3 only).
6. κ is a Monte Carlo estimate (13 % spread) on the logic netlist, 5,160 decisions per block; the timing-closed lane's balancing buffers and stations are decisions too and are not emulated, so κ understates its error power by an uncomputed factor.
7. The 300 K power is a hydrodynamic-tier τ^{-3} scaling [D3] with no kinetic correction applied, and every correction the record evaluates would raise it — the closure's 7–16 % higher 300 K threshold, and, in graphene's own fluid, a tracked bias costing ×1.2–3.2 the stored 353 K cell dissipation wherever a threshold exists (notes 2026-07-13, 2026-10-15; Section 4.5); the 69-GHz slot is unmodelled at 300 K; and cold-side physics — lifetime saturation, the Joule model — is uncertified below the ≈255 K closure line.
8. Biased cells burn power when idle: lanes 51–88 % busy dissipate as if full, and the fJ-per-MAC figures assume full utilization.
9. The yardstick is a proxy; the model's tolerance of lane errors is unmeasured.
10. The inputs are synthetic: Bonsai-shaped weights with an assumed one-third share of zeros (external: measured zero shares reach up to 51.5 % across 29 ternary models [33]), and Gaussian or Student-t activations rather than Bonsai's own rotated ones; the Q8_0 activation format is itself an assumption (Section 3.1).
11. Memory, interface, launch and CMOS-perimeter energies are not booked; the wall-plug launch energy depends on a launch-coupling efficiency the record does not quantify (Part I §9.5), and no note prices the clock distribution — power per launch point, number of launch points, master-to-station loss (issue #139) — or the energy per bit of an electrical write and read (issue #141).
12. The ≈800 lines at 69 Gb/s — modulation, phase and coupling into the acoustic plasmon — are not modelled, nor is a cryostat interface for them, nor a readout's own decision: the record prices decisions only at deciding-cell inputs, where even a homodyne-limited decision at DG1's pass line needs ≥ 406 quanta at 300 K (477 at 353 K) against the 99–277 a loaded F = 2 cascade sustains (in-model; note 2026-10-08; Section 4.6). An electrical write and read through a cell's gate and its own rectification has its couplings priced (note 2026-10-16; the gate response in-model, the rectification demonstrated), but not the gate's RC, the slot-rate electronics or the read budget (issue #141).
13. The rail plane decides analog feasibility and is unanchored: QFA-3's 4 K error differs 30-fold between P1 and P2; and in the released model no cell output saturates below near-depletion, so the rail is a coded cap at whichever plane is chosen, not a clamp the cell supplies (note 2026-10-08; demonstrated in the released model, which leaves its physical domain above a few-percent drive).

14. N_op = 2,600 is 6.5× the registered 400, whose "launch-budget" cap the record never derives: no released key or manuscript sentence defines what it denominates, and 400 quanta is 18.41 dB over $\bar{n}$ at 300 K counted in quanta but 21.06 dB in quadrature (note 2026-08-25 §1).
15. The relaunch model [E7] is minimal: a '1' at the rail and a '0' as the released amplifier's unsaturated output, both attenuated to N_op. No run of the record delivers a rail-scale '1' to a deciding input, and the released chain has no cell-to-cell signal-transfer model; if the relaunched '1' is instead the loaded F = 2 fixed point (in-model; 111–277 quanta on the absolute bias leg, 99–255 on the ratio leg), N_op under the same attenuator is 14–35× (15–39×) smaller than 2,600, and the REF and CONS errors move the worse way (not computed; the registered 400 already gives 2.3×10^{-2} at 4 K; note 2026-10-08). Its '0' carries no memory of the preceding bit, whereas in the digital fabric's gated slot the released solver leaves a coherent residue in the '0' that follows a '1' (−15.9 to −15.7 dB below the mean '1' and 82–86° off its phase at 67 GHz on N = 240, 65–69° at N = 720; note 2026-08-12, demonstrated at the model's ratio bias and released time step); its size at the tile's temperature, slot and plane is not computed, and carrying it into the quantum chain is a chain-crossing step no promoted note establishes.
16. Junction and split conventions [E1, E2] are choices, and the sensitivity to −3 dB junctions carries the second check's caveat.
17. The comparator offset c is open: $1/\sqrt{12}$ in REF and CONS, 0.02 in PERM.
18. The launch-noise convention is Part II's additive one, referenced to the rail swing at the first decision for the digital lane ([D8]) and to the N_op plane for QFA-3; the record's four launcher conventions span ×239 in Part II's 2-bit decode error at 4 K and N_op = 400 (note 2026-08-25, launcher-physics fork, check P3; Section 5.4), and the plane fork also moves the station spacing (Section 8.1(f)).
19. The $k \geq 16$ comparator redesign, a margin choice Part II makes for its three-level ladder and carried here to four levels without re-derivation, is neither designed nor charged.
20. The booking fork (A0 or B) is unresolved; the analog verdict is computed under A0 and argued under B, where the listing does not recompute the hybrid: the digital lane must itself run below about 60 K, so the comparison is again between two cold lanes and their coolers, its numbers not computed (Section 6.6).
21. Cell-accounting variants are estimates, and the analog listing's all-digital lane (5,342–5,641 cells, its logic omitting the 32 sign-bit inverters) differs from the timing-closed lane (8,169–8,455).
22. The threshold full adder, its three-input fan-in and its weighted inhibitory coupling are untested; in the amplitude encoding the released cell supplies, in-model, neither a threshold on a two-input sum (2.6–3.3 dB of separation) nor, for carrier-frequency inputs, a depleting coupling (note 2026-10-16).
23. The model-side levers are projections on synthetic data; no fine-tuning was run.
24. Phase-sensitive amplification (Part II §7) is explored only in-model, outside this Part's listings, and enters none of this Part's computed numbers: on the released Galilean cell a $2f_0$ pump at 5 % of the source density gives 8.7–9.3 dB of phase-sensitive contrast (drain velocity, signal 2×10^{-3}, ratio bias leg, N = 720–240) and restores phase only in narrow, grid-scoped windows; above an onset the cell latches, and gated in note 2026-07-31's 69-GHz slot at one drain pump the latch amplifies a bit stream's flips but re-levels none; its noise (the evasion of the noise factor $2 - 1/G$) is not computed (note 2026-10-17; Section 8.1(g)). Not explored: squeezing; a CMOS per-slot sampler; subranging decoders; wavelength-division or optical operand fan-out; and a dual-rail binary lane (each signal carried with its complement, not Section 6.1's dual-rail ternary), in which NOT is a wire swap and the lane's 783 inverting cells (Section 4.2; note 2026-10-13's count) become wiring, at the cost of complementing every gate. It removes the NOT, not the AND, which the released cell does not supply as a threshold either; of the inverter options note 2026-10-16 lists, it is the one the record can price without new physics, and its count was not run.

25. The 69-GHz slot rests on a floor specification, a floor observable and a bias convention the record has not fixed (notes 2026-08-01, 2026-08-02). The specification cannot be derived from the released chain's compact logic map (cell.transfer) and must be imported, from the bench or from a downstream decode demand, a demand itself set by conventions the released code leaves open — comparator re-trim, the phase projection, k_dec and c (note 2026-08-02's appended assessment record, reproduced in note 2026-08-12) — and the fabric-side '0' residue any such specification is applied to is a property of the released scheme at its released time step (at 67 GHz and N = 240 the slot-mode residue after a '1' moves 10.3 dB across the solver's time-step factor 0.2–0.8; note 2026-08-12, demonstrated). The bias is pinned only bench-side, as a proposal (0.7 × the measured threshold; note 2026-08-01, bench gate G1 reissued); in the solver the fork persists into the continuum (≈ +0.3 dB of gain at 67 GHz; in-model), and the requirement leg of the bias-consistent comparison still carries an imported operating amplitude (note 2026-08-01, ratio-bias gain table).
26. The ±1 dB contact allowance, possibly larger for a diffuse contact, is open and gated by L0 (Part I §7.5, §10).
27. Read at the deciding cell's input, the only plane at which it pairs with input-referred noise (note 2026-10-08), a rail-scale swing is a ≈10 % source drive. There the released solver's Galilean fluid, which has no velocity cap, runs at 1.4–1.6 times graphene's Fermi velocity v_F, which it exceeds above a ≈0.03 drive. What the released cell does at rail-scale drive is therefore a statement about the released model, and the physical cell is expected to leave that model's domain above a few-percent drive (in-model). The released chain also has no cell-to-cell signal-transfer model (coupled.py is a threshold and stability model), so the '1' a deciding cell receives is a ledger reading, not a run (note 2026-10-08).
28. The tracked bias presupposes a threshold. The record's hydrodynamic tier and note 2026-07-13's kinetic closure have one; in that note's §6 single-rate finite-temperature kinetic operator there is none below the Landau ceiling at 353 K at the central and lower electron–electron rates of its band when the electrons shed their drift heat, and at 300 K one only at the band's top; whether graphene's own electron fluid has one at 353 K turns on the open rate and finite-drift structure of the relaxation of the current not locked to momentum: three admissible two-rate continuations, identical at zero drift, give three answers (in-model; note 2026-10-15; Section 2.1). Every physical number here presupposes it.
29. The lane's 783 inverting cells and its ANDs have no mechanism of Part I's Eq. (5) kind in the released cell (in-model; note 2026-10-16; Section 4.2). In the amplitude encoding an inverter exists only by destructive interference with a coherent clock pulse held within 10.1–12.3° of phase (by grid, tightening under refinement; restoration open in the continuum), or behind a biased detector segment and an external ×9–12 gain stage, whose rectified shift carries 21–35 % of itself into the next slot and whose gate RC is open; the ANDs separate '1 + 1' from '1' by only 2.6–3.3 dB. Under phase encoding a NOT is a half-period delay, with restoration from a $2f_0$ pump admissible in-model only in a narrow window (note 2026-10-17); the dual-rail count is not run (limitation 24).
30. Every amplifier step in the listings — QFA-3's relaunched '0' and its decoder noise, booking B, and the decision noise of bookings A0–A3 — uses the released amplifier rule with its idler at the lattice occupation $\bar{n}(T)$. Part II's cold 2-bit cut (QMAC-1, N_op = 400, quiet launch; the released 58.5 K) is a statement about that idler: with the idler pinned, 10^{-3} is reached below 105.7 K for a vacuum idler, below 45.8 K at one idler quantum and at no evaluable temperature from 1.5 quanta (note 2026-08-05 §4; demonstrated), so for an idler no colder than the lattice Part II's table is optimistic-or-exact on the idler axis, with a lever of opposite sign on the gain booking unsettled (its §6). QFA-3's cold envelope assumes the idler cools with the lattice; its sensitivity to a hotter idler is not computed (gates TG1 and TG3 would see it).

31. No carrier-phase budget is carried for the lane. The comb fixes the carrier phase at the launch (Part I §3.2); in-model the cell's own transfer then moves it 12–21° per 1 % of drive detuning and 4–6° per 0.05·M_th of bias (note 2026-10-13), against Part I's ±15° clock budget. What the lane can spend rests on two open forks. In the clocked regime of Part I's Eq. (5) every cell amplifies the pump, and the requirement is one cell's: −0.85 / +1.15 % of f_0 (shipped ratio-leg continuum). If two-input gates sum fields in a station-free cascade, a between-branch error must stay within ±0.32 to ±0.010 % of f_0 per cell over 3–91 non-shared cells; if gates couple envelopes, only the latency matters (note 2026-10-13). The field-sum branch has no in-model threshold AND (2.6–3.3 dB of separation), in the amplitude encoding a cascaded datapath with no clock fed in has no inverter, and every in-model inverter is pump-fed by the clock burst, the interferometric one inside a 10.1–12.3° input-to-pump window, by grid (note 2026-10-16). A $2f_0$-pumped cell would hold a per-cell error to a fixed offset up to a 19.2° slip margin, in a narrow window (note 2026-10-17; Section 8.1(g)). The regime and the gate model (field sum or envelope coupling) are open (Section 2.2).

10. Conclusion

Part I asked whether room-temperature graphene can gain back what a plasmon loses, and Part II what the fabric is worth when its pulses are counted in quanta. This Part asked what the fabric is for, given a demanding workload.

What maps is one inner loop, exactly. A 32-wide block sum of a ternary model's signed 8-bit products fits on a lane of Part-I cells that is exact by construction — demonstrated as a netlist on 1.70 M Bonsai-shaped block sums — and, in-model, takes 8,169–8,455 cells once timed for latch-free operation. What the fabric is for is energy and area at memory speed, not speed: two to seven lanes keep pace with 0.5–3 TB/s in at most about 1.7 mm^2 and 0.64 W at 300 K (hydrodynamic tier), the decode stays as fast as the memory and no faster, and the hardest engineering is the ≈800 operand lines that a fabric without storage needs.

What decides is one measurement. The lane is correct as physics only if a loaded cell decides as well as the released booking says, at the swing its deciding cells actually receive: that booking holds on the coded rail swing, which no run of the record delivers to a deciding input, and at the swings the record's own loaded cascade sustains no booking passes warm (in-model; note 2026-10-08). On the rail swing, the split and junction losses of Part I's fan-out budget (Part I §6.3) would make it a 144 K machine for the −3 dB split alone and a 114 K one through a −1 dB junction (colder at −3 dB), Part II's calculus a cryogenic one, and DG1, reported with the input-referred swing, chooses. Interference counters help only when cold, save at most about a third of the cells and leave the comparison to the cooler: QFA-3 is a test tile, not a route. The levers lie mostly outside the fabric — 4-bit activations (near parity reported externally with quantization-aware training; Section 8.2) or error-aware fine-tuning would raise the rail-swing pass temperatures by tens of kelvin, and a threshold full adder would halve the lane's logic if the cell thresholds a three-way sum and takes a depleting input; in the amplitude encoding the released cell, in-model, separates two in-phase inputs from one by only 2.6–3.3 dB and has no depleting input for a carrier-frequency signal, and a three-way sum is untested (note 2026-10-16).

None of the in-model statements is yet a measurement, the gain cell beneath all of them is unproven, and whether graphene's own electron fluid has the drift threshold that cell is biased against at 353 K is open (note 2026-10-15). The design is now exactly as falsifiable as Parts I and II were built to be.

Appendix TA. The listings and their correction histories

The four listings live in fable-model-ternary/ (Table TA1). Each imports the released chains read-only (nothing edited, no bytecode written, run_all.py never run) and prints every number it computes for its

sections (values from the community record enter as cited inputs); the .out file beside each is its reference output (macOS arm64, Python 3.11.2, numpy 2.4.6; runtimes in Table TA1). A re-run of 24 October 2026 on Windows 11 (x86-64, Python 3.11.9, numpy 2.4.6) reproduces every printed digit of the three computing listings except the runtimes the digital-lane and variants listings print (the per-matrix emulation times and the totals, 56 s and 46 s on that machine against Table TA1's 21 s and 16 s on the reference machine) and the last floating-point digit of five full-precision keys of ternary_qmac_design.py §7; every rounded value of Table TA2 is identical. digital_lane_bonsai.py §2 prints its unit-delay conversion (≈31 ps at ≈0.33 ps per cell, 2.1 slots on the 92-cell path), a unit note 2026-10-13 contradicts in-model (an isolated gated 12-ps burst is delayed 2.7–4.0 ps per cell by its energy centroid and 2.0–7.3 ps across envelope estimators on the released grids; 13–37 slots in flight on the 92-cell path across the rise-to-centroid and narrowband estimators); Sections 4.3 and 9 quote the note. The cell counts do not depend on the unit, and the note's mirror of the depth model reproduces them element-wise.

digital_lane_bonsai.py and ternary_qmac_design.py, the two design studies folded into Sections 3–7, are at their second revision (v2); before it, an independent adversarial verification agent re-checked each, and the findings were adopted. The digital-lane check found four material defects — unbooked wave-pipeline timing (now [D7]), the released booking shown only unloaded (now A1–A3), missing launch noise (now [D8]) and 300 K power paired with a slot rate adjudicated at 353 K (now stated), together with a superseded hydrodynamic 353 K low end (now kinetic-corrected) — and seven minor ones. The analog check found five — equal cold and warm cell dissipation (now [E8]), a misattributed rail plane (now P1 and P2, [E4]), extra junction levels without their quarter-wave, missing relaunched '0' noise (now [E7]) and uneven cell accounting — and seven minor ones. A second check of the analog listing raised five more: an unphysical gain at −3 dB junctions in the rail re-solve (N_op re-solved to 6,530.5 quanta, 1.70 N_rail, so the relaunch attenuator runs as a +2.3-dB gain), the hybrid alignment accounting, a 1.7-dB SNR overstatement from per-block scales (Student-t; 0.2 dB for Gaussian), the TG1 threshold and the warm reference temperature. They are open rework items for a third revision that is on hold, and Section 6's numbers carry that caveat; the second check is not stored, and no listing prints its estimates. The analog listing's all-digital reference also omits the 32 sign-bit inverters (Section 6.4). The digital listing has since become v2.1, which also prints both ends of the heat range, the booking-B mapping factor (KAPPA_B_RATIO) and the mapped values; no computed quantity changed.

variants.py and figures.py were written for this manuscript. figures.py transcribes listing outputs, except that it computes the A-booking curves of Figure T3A from the released chains and maps the B points with the 300 K factor. The extensions each listing adds to the record, and the digital listing's external inputs [D6], are defined one per line in Table TA3. The listings pre-register new result keys (Table TA2); none collides with either released results.json, and none contests a stored value.

File	Role	Extensions	Used in	Runtime
digital_lane_bonsai.py (v2.1)	the Part-I digital lane: workload, netlist, timing closure, bit-exact emulation, κ and the pass line, bookings A0–A3 and B with the booking-B mapping factor, launch noise, system, heat, CMOS perimeter	[D1]–[D8]	§2–8	21 s
ternary_qmac_design.py (v2)	the Part-II analog counter tree: K-input tile and its bit-for-bit check, split law, ternary encodings, QFA-3 envelope and sensitivities, hybrid lane, cooling, booking fork	[E1]–[E8]	§2, 4–7	2 s
variants.py	the levers: lane at b activation bits and block width W, threshold full adder, yardstick and ceiling at b bits, pass temperatures, tolerance	[V1]–[V5]	§8	16 s
figures.py	Figures T1–T5	—	all	—

Table TA1. The listings in fable-model-ternary/. Each prints every number it computes for its sections; the .out files are the reference outputs on macOS arm64 (Python 3.11.2, numpy 2.4.6), and the runtimes are that machine's.

Key	Value	Listing
dl_lane_cells_t0_low / _high	8,169 / 8,455	digital_lane_bonsai.py §6
dl_balancing_buffers_t0	2,616	digital_lane_bonsai.py §6
dl_kappa_int2_per_block_per_BER	2.82×10^{8}	digital_lane_bonsai.py §6
dl_ber_ceiling_equal_q80_t4	6.28×10^{-9}	digital_lane_bonsai.py §6
dl_snr_A0_300K_t4_dB / dl_snr_A1_300K_t4_dB	64.9 / 40.6	digital_lane_bonsai.py §6
dl_snr_B_300K_dB	−4.0	digital_lane_bonsai.py §6
dl_kappa_B_ratio_300K	0.051	digital_lane_bonsai.py §6
dl_launch_S_min_300K_k8_dB	27.9	digital_lane_bonsai.py §6
dl_lanes_for_3TBs	7	digital_lane_bonsai.py §6
qfa3_err_4K / 20K / 77K_p2_relaunch_c0289	1.302×10^{-5} / 3.917×10^{-5} / 5.016×10^{-3}	ternary_qmac_design.py §7
qfa3_err_4K_p1_relaunch_c0289	4.010×10^{-4}	ternary_qmac_design.py §7
qfa3_err_4K_p2_quiet_c002_multiport	4.058×10^{-13}	ternary_qmac_design.py §7
qfa3_N_op_p2_relaunch_cold	2,599.8	ternary_qmac_design.py §7
qfa3_T_below_1e-4_p2_relaunch_c0289_K	27.17	ternary_qmac_design.py §7
relaunch_zero_sigma_max_pct_4K	1.558	ternary_qmac_design.py §7
tq_allqfa_snr_dB_4K_p2_relaunch_t4	28.23	ternary_qmac_design.py §7
tq_bstar_77K_p2_relaunch_t4	2	ternary_qmac_design.py §7
tq77_carnot_fraction_needed_joule	0.396	ternary_qmac_design.py §7
ha_decode_error_bookingB_300K	5.479×10^{-3}	ternary_qmac_design.py §7
p3_counters_b8_W32	236	variants.py §4
p3_logic_cells_b8_W32_record / _b4_W32_record	5,160 / 2,532	variants.py §4
p3_logic_cells_b8_W32_TFA / _b4_W32_TFA_selectCMOS	2,680 / 620	variants.py §4
p3_ceiling_b8_t4 / p3_ceiling_b4_t4	6.05×10^{-9} / 1.25×10^{-6}	variants.py §4
p3_yardstick_b4_t4_dB	17.9	variants.py §4
p3_T_pass_A1_b8_K / p3_T_pass_A1_b4_K	291.3 / 400 (top of the scanned range)	variants.py §4
p3_T_pass_A2_tol10_K	167.6	variants.py §4

Table TA2. Pre-registered keys (new names, rounded to the precision the estimators support). None collides with fable-model-chain/ or fable-model-quantum/ results.json, and none contests a stored value.

Label	Definition (one sentence, from the listing header)	Listing
[D1]	The lane netlist: 32 int8 × ternary products per slot, select (p ∧ a) ∨ (n ∧ ¬a), sign bit inverted, a greedy Wallace tree of record full adders and a 12-bit ripple adder, with the block constants and scales applied in CMOS.	digital_lane_bonsai.py
[D2]	Gate-level emulation in which every cell decision flips independently with probability BER.	digital_lane_bonsai.py
[D3]	τ^{-3} scaling of the stored 353 K cell dissipation to 300 K at the tracked bias (hydrodynamic tier); no kinetic correction is applied, and every correction the record evaluates would raise it (notes 07-13, 10-15).	digital_lane_bonsai.py
[D4]	Activation broadcast to L lanes through F = 2 buffer trees.	digital_lane_bonsai.py
[D5]	Bookings A1–A3 (the released margin law with the F = 2 softening, then the split and junction attenuation of Part I's worst path) and B (the Part-II calculus), beside the released booking A0; the split-plus-junction charge is that of Part I §6.3's fan-out budget, and all five are referred to the coded rail swing (note 10-08).	digital_lane_bonsai.py
[D6]	The model's workload constants, external inputs from the public configuration and file sizes.	digital_lane_bonsai.py
[D7]	Timing: a depth model of the netlist in which every live wire is regenerated at every level (balancing buffers, skew tolerance t) and a re-sync station sits on every wire crossing a cut at every station spacing.	digital_lane_bonsai.py
[D8]	Launch noise: Part II's launch excess variance added at the first decision after each launch; referenced to the rail swing X_RAIL, so at a smaller delivered swing the errors would be larger (note 10-08; direction only).	digital_lane_bonsai.py
[E1]	K-input combiner: a balanced binary tree of the released combiner, every arm passing the same extra junction levels, each with a quarter-wave of propagation.	ternary_qmac_design.py

Label	Definition (one sentence, from the listing header)	Listing
[E2]	Comparator fan-out to K comparators: 3.0 dB per halving ($\log_2 K$) plus $\lceil \log_2 K \rceil$ junction levels, each with its quarter-wave; K = 2 is the released value exactly.	ternary_qmac_design.py
[E3]	Decision: released midpoint thresholds, worst-case symbol variance, decoder amplifier noise and the static offset term $V_{static} = c^2(x_{range}/k_{dec})^2$.	ternary_qmac_design.py
[E4]	Operating point: N_op capped by N_rail at a stated plane, P2 (top level at the combiner output) or P1 (the pre-loss launch sum).	ternary_qmac_design.py
[E5]	Cells: K comparators, thermometer-to-binary logic and alignment buffers in the record's idiom, with accounting variants carried explicitly.	ternary_qmac_design.py
[E6]	Signed tile: phase-encoded trit arms (phase π = −1, gated = 0) plus a coherent offset arm at combiner weight ½.	ternary_qmac_design.py
[E7]	Relaunch: inside a tree a '1' is a rail pulse and a '0' a cell's unsaturated output, both attenuated to N_op before the released input transit; no run of the record delivers that rail pulse (note 10-08).	ternary_qmac_design.py
[E8]	Cell dissipation by the released Joule form at the tracked bias, scaling as τ^{-3}; uncertified below ≈255 K; graphene's own Joule dissipation $J \cdot E_0$ exceeds it ×1.25–1.35 wherever a tracked threshold exists; at 353 K whether one exists is open (note 10-15).	ternary_qmac_design.py
[V1]	The lane generalized to b activation bits and block width W in the same netlist idiom.	variants.py
[V2]	A native threshold full adder: carry as one majority cell, sum as one cell with a weight-2 depleting coupling from the carry, plus three alignment buffers (5 cells, depth 2); in the amplitude encoding the released cell supplies, in-model, neither its threshold on summed fields nor its depleting coupling (note 10-16).	variants.py
[V3]	The yardstick at b bits: the same absmax-per-32 quantizer at b bits (Q8_0 at b = 8), K = 5,120, Student-t and Gaussian activations, one-third zero weights.	variants.py
[V4]	κ_b by gate-level emulation with injected decision flips at b bits, and the per-decision ceiling that follows.	variants.py
[V5]	The tolerance lever: pass temperatures when the model tolerates fabric error power 1×, 10× or 100× its own quantization error.	variants.py

Table TA3. The extensions named in brackets throughout, one per line, as defined in the listing headers, with the record's qualifications after a semicolon; [D6] is an external input, listed for completeness, not an in-model extension. Everything not labelled here is a released call of the record's chains.

Appendix TB. Symbols

Symbol	Meaning	Value or range
w; a	ternary weight; int8 activation (Q8_0)	{−1, 0, +1}; 8 bits
p, n	weight bit planes [w = +1], [w = −1]; p is also the QFA-3 symbol error (§6) and n the carrier density (§6.5)	one wire each
x_m; u; U	product bit m; unsigned product; block sum, eq. (T1)	$u \in [0, 255]$; $0 \le U \le 8{,}160$
n_neg	number of −1 weights in a block	0–32
i_1, i_2, c_in; s, c_out	full-adder input bits and carry in; sum and carry out, eq. (T2)	bits
b; W	activation bits; block width	8; 32 (reference)
L	lanes	2 / 3 / 7
K	dot-product length (§5); counter inputs (§6)	5,120; 3 for QFA-3
N_B	blocks per K = 5,120 dot product	160
κ, κ_b	block-sum error variance per unit decision BER (at b bits)	2.8×10^8 int^2 (κ, reference lane); κ_b = 2.93×10^8 at b = 8 (variants.py), within κ's 13 % spread
SNR_q; P_sig	yardstick (Q8_0 dot-product SNR); signal power in lane units, eq. (T3)	42.5 dB (Student-t), 45.5 dB (Gaussian)
BER_ceil	pass line per cell decision, eq. (T3)	6.3×10^{-9} (2.6×10^{-6} per launched bit)
k; k_0; k_eff	threshold sharpness β·A_swing/Δn (Part I); the released code's design constant for it (cell.py); at fan-out 2	8; 8; $8/\sqrt{2} = 5.66$ (design rule); released cell 0.18–0.26 by density coupling, ≈2.9 by interference (tanh scale; note 10-16)
F	fan-out	2
NF	cell noise figure (Part I's F)	2.77-dB floor; 4 dB (Part I's requirement)

Symbol	Meaning	Value or range
S; σ_rel	launch SNR; launcher amplitude jitter σ_A/Ā of the launched '1' (σ_rel = 10^(−S/20) in Part II's coding; Part I's 20 dB is a link budget against the thermal floor, note 08-25)	coded 20 dB = 10 %; needed 28–33 dB
N_op; N_rail	quanta per input pulse; rail quanta	2,600 (REF), 1,278 (CONS), 1,822 (PERM), 400 (registered); 3,833
n̄; T_Q	thermal occupation of the mode; quantum crossover $\hbar\omega_0/k_B$	5.8 at 300 K; 48 K
f_0; $2f_0$	carrier frequency of the cell mode; a pump at twice it (§8.1(g))	1 THz; 2 THz
c	comparator rms trim offset, fraction of transition width	$1/\sqrt{12}$ (REF, CONS); 0.02 (PERM)
σ_0	noise of a '0' level (relaunched), fraction of the rail swing	3.56 % released at 4 K; ≤ 1.56 % needed
P1, P2	rail reference planes: pre-loss launch sum; combiner output	CONS; REF, PERM
Δ; η_in, η_extra, η_fo	per-comparator level spacing; transmissions, eq. (T4)	$\Delta^2 \propto K^{-3}$ (roughly) at P2
p; b∗; j	QFA-3 symbol error (p is also a weight bit plane, §4); highest analog bit column; bit column	1.3×10^{-5} at 4 K (REF); 6 at 4 K
r; η_c; T_a	cold/warm cell dissipation ratio; cooler efficiency (fraction of Carnot); ambient, eq. (T5)	0.125 (Joule model); —; 300 K
P_wall, P_warm; N_cold, N_warm; P_cell(T)	wall-plug power of a cold lane and power of the warm all-digital lane; their cell counts; dissipation per cell at T, eq. (T5)	P_wall/P_warm 0.47–8.4 (Joule model, ideal cooler); P_cell 7.78–10.22 μW (300 K), 0.97 μW (≤ 150 K, from the stored value; r = 0.125 at either end)
n; m∗; v_0	carrier density; effective mass E_F/v_F^2; drift velocity (Part I; Joule form, §6.5)	10^{12} cm^{-2}; —; ≈2.4×10^5 m/s at 353 K (Part I Table 3; tracked bias)
M, M_th	drift Mach number; its DS threshold (Part I)	M/M_th = 0.7 (tracked)
τ	plasmon momentum-relaxation lifetime	1.0 ps (300 K), 0.85 ps (353 K), 2 ps saturated (≤ 150 K); Part I's best-case row (Part I §4; issue #140)

Table TB1. Symbols of Part III. Part-I symbols M, M_th, τ, k (= β·A_swing/Δn), n (carrier density), m∗ and v_0 carry over. Part I's plasmon speed s and cell length L are not used (s here is the full-adder sum, L the number of lanes), and Part I's noise figure F is written NF, since F denotes fan-out. K, p and n each carry two meanings, kept apart by section.

Data and code availability

The four listings are released in fable-model-ternary/ of the repository of Parts I and II (https://github.com/ryoji-info/FableComputer), beside the released chains they import read-only. Every number this Part computes reproduces from the four scripts (Python 3 and numpy; matplotlib for the figures), with the reference outputs beside them; the second check's rework estimates quoted in Section 6.4 and Appendix TA are not stored, and no listing prints them. Values taken from the community record [12] — the 69-GHz frontier of note 2026-07-31, for example — enter the scripts as cited inputs or are quoted in the text, and reproduce from each note's own listings where it carries or links them, otherwise as the note states. Values from Parts I and II and from external sources carry their citations — the runtime activation formats of Section 3.1 come from PrismML's code, read on 24 October 2026, and enter no listing — and in-text derivations are shown where they appear. Items tracked as repository issues [25] are cited as open, not as results: the linewidth measurement behind gate L0 (a measurement request, issue #88), the staged bench plan (a call for an experimental partner, issue #97) and three design questions, the module's clock source, the material gap and the CMOS boundary (issues #139–#141). The revision history is kept in papers/REVISION-HISTORY.md.

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